MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
6000 VILLAGE: CLASS-B ACCOMMODATION
DESIGN OF BUILDING STRUCTURE

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
Typ.	Typical
HCS	Hollow Core Slab
Fc'	Concrete Compressive Strength
Fy	Yield Strength of Reinforcing Steel
ϕ	Strength Reduction Factor
SBC	Saudi Building Code
SASO	Saudi Standards, Metrology and Quality Organization
ACI	American Concrete Institute
ASCE	American Society of Civil Engineers
PCI	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of an Accommodation B Building.

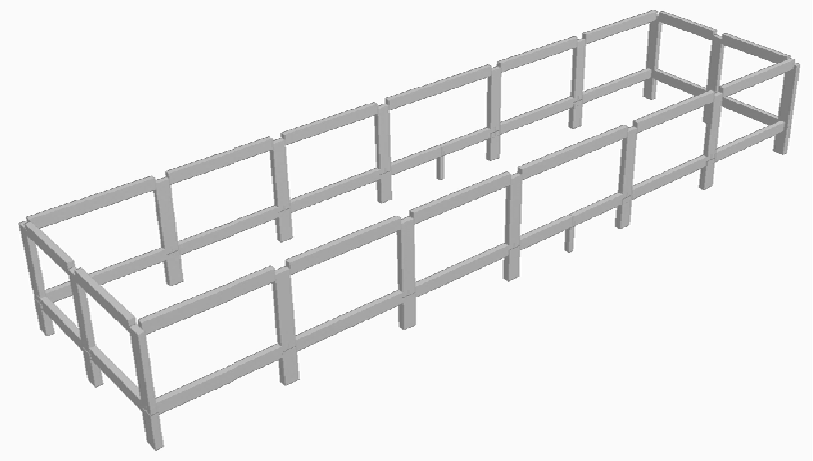
2. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Accommodation B Building only.

3. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HSC (Hollow Core Slab)

The Superstructure for this Building was modelled using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base that are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.



STAAD 3D MODEL

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3.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

3.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008.

3.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel.

4. DESIGN LOADS

4.1. Dead Load

4.1.1. Roof Dead Loads

- Provision for Electrical Lighting = 0.05 kN/m²

Ceiling

Suspended steel channel system = 0.10 kN/m²

Acoustical Fiber Tile (16mm) = 0.05 kN/m²

Sub-total = 0.2 kN/m²

- Hollow Core Slab

(HCS-250)(To be confirmed by Vendor) = **3.43 KN/m2**

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Prestressed Hollowcore Floor

Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m² for Finishes Characteristic Service Loads Kn/m²							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/m²**

Total Load =3.43+0.2+3.1= 6.73 **kN/m²**

UDL on Edge Ledger Beams (LB1&LB2) = 6.73 x 8.75/2 = **29.45 Kn/m**

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+ (0.05+0.2)/2x20)*0.6*0.6/2 = **1 knm/m**

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4.1.2. Ground Floor Walls

- 250 tyke. Outer CMU (Partially Grouted Cell@800 c-c)

16 x 0.25 x (3.5-0.5) m clear height = **12 kN/m**

4.2. Live Load

4.2.1. Roof Live Loads

Live loads are with reference to Clause-4.2.2, MD-216-1000-EG-ST-CRT-1001

Ordinary Flat (Inaccessible) = **0.5 kN/m²**

UDL Due to Live Load on the Edge Ledger Beam (LB1& LB2) =0.5X8.75/2= **2.1875 kN/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 0.5*0.6*0.6/2 = **0.09 kN/m /m**

4.3. Wind Load

Wind loads were automatically generated in STAAD as per ASCE-07. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

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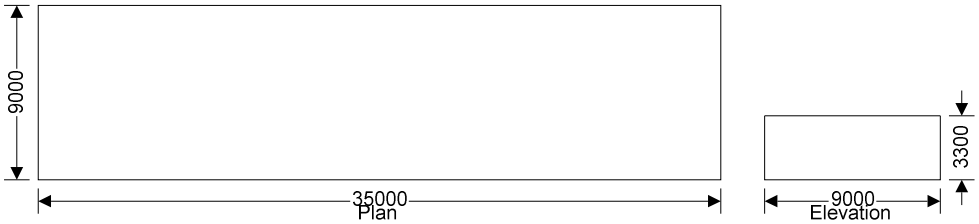
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4.4. Wind Load (Analytical Verification)

WIND LOADING (SBC-301)

Using the directional design method



Building data

Type of roof;	Flat
Length of building;	b = 35000 mm
Width of building;	d = 9000 mm
Height to eaves;	H = **3300** mm
Mean height;	h = **3300** mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff ;	K_d = **0.85**(As per SBC-301, Table 6.4-1)
Exposure category;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification;	Partially enclosed buildings(As per SBC-301, Section-6.2)
Internal pressure coef +ve;	GC_{pi_p} = **0.55**(As per SBC-301, Figure-7.2-1)
Internal pressure coef –ve;	GC_{pi_n} = **-0.55**(As per SBC-301, Figure-7.2-1)
Gust effect factor;	G_f = **0.85**(As per SBC-301, Section-7.2.7.1)
Importance Factor;	1(As per SBC-301, Section-9.5)

Topography

Topography factor not significant;	K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)
Velocity pressure equation;	$q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2 xI$ V in Km/hr(As per SBC-301, Section-7.2.8)

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Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m^2)
3300	1.03	1.29
3301	1.03	1.29

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	0.17	115.53	19.35
B	3.30	-0.50	1.29	-1.26	115.50	-145.11
C	3.30	-0.70	1.29	-1.48	29.70	-43.82
D	3.30	-0.70	1.29	-1.48	29.70	-43.82

Overall loading

Projected vertical plan area of wall; $A_{vert_w_0} = b \times H = 115.50$ m^2

Projected vertical area of roof; $A_{vert_r_0} = 0.00$ m^2

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0} =$ 88.9 kN

Leeward net force; $F_l = F_{w,wB} = -145.1$ kN

Windward net force; $F_w = F_{w,wA} = 19.4$ kN

Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 164.5$ kN

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Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	1.58	115.53	183.12
B	3.30	-0.50	1.29	0.16	115.50	18.60
C	3.30	-0.70	1.29	-0.06	29.70	-1.72
D	3.30	-0.70	1.29	-0.06	29.70	-1.72

Overall loading

Projected vertical plan area of wall; $A_{vert_w_0} = b \times H = 115.50\ m^2$

Projected vertical area of roof; $A_{vert_r_0} = 0.00\ m^2$

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0} = 88.9$ kN

Leeward net force; $F_l = F_{w,wB} = 18.6$ kN

Windward net force; $F_w = F_{w,wA} = 183.1$ kN

Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 164.5$ kN

Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	0.17	29.71	4.98
B	3.30	-0.21	1.29	-0.93	29.70	-27.74
C	3.30	-0.70	1.29	-1.48	115.50	-170.41
D	3.30	-0.70	1.29	-1.48	115.50	-170.41

Overall loading

Projected vertical plan area of wall; $A_{vert_w_90} = d \times H = 29.70\ m^2$

Projected vertical area of roof; $A_{vert_r_90} = 0.00\ m^2$

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_90} + p_{min_r} \times A_{vert_r_90} = 22.9$ kN

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Leeward net force; $F_l = F_{w,wB} = -27.7$ kN

Windward net force; $F_w = F_{w,wA} = 5.0$ kN

Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 32.7$ kN

Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	1.58	29.71	47.09
B	3.30	-0.21	1.29	0.48	29.70	14.36
C	3.30	-0.70	1.29	-0.06	115.50	-6.70
D	3.30	-0.70	1.29	-0.06	115.50	-6.70

Overall loading

Projected vertical plan area of wall; $A_{vert_w_90} = d \times H = 29.70$ m^2

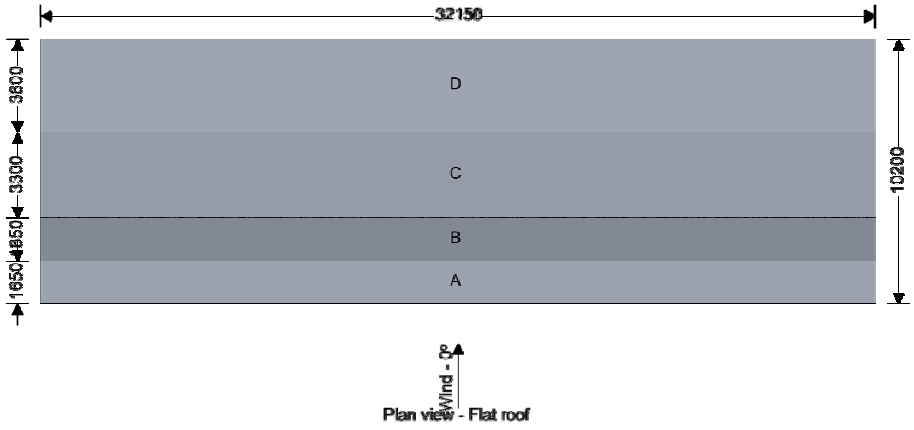
Projected vertical area of roof; $A_{vert_r_90} = 0.00$ m^2

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_90} + p_{min_r} \times A_{vert_r_90} = 22.9$ kN

Leeward net force; $F_l = F_{w,wB} = 14.4$ kN

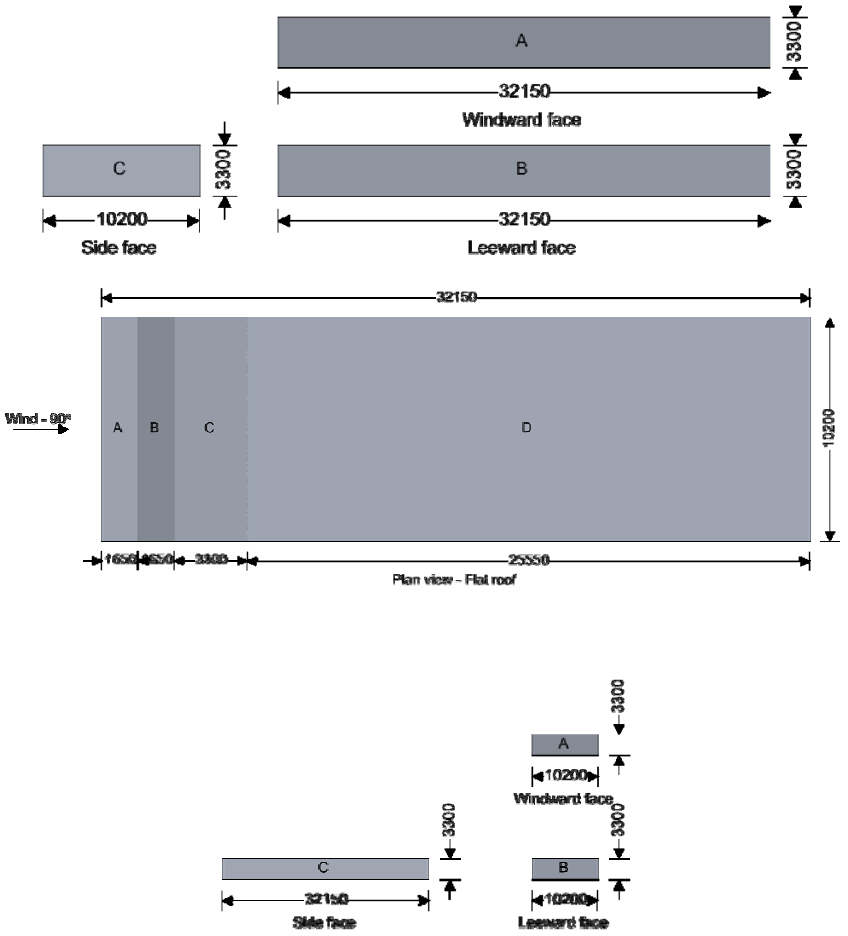
Windward net force; $F_w = F_{w,wA} = 47.1$ kN

Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 32.7$ kN



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So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

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4.5. Seismic Load

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST-CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

4.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

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5. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

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6. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

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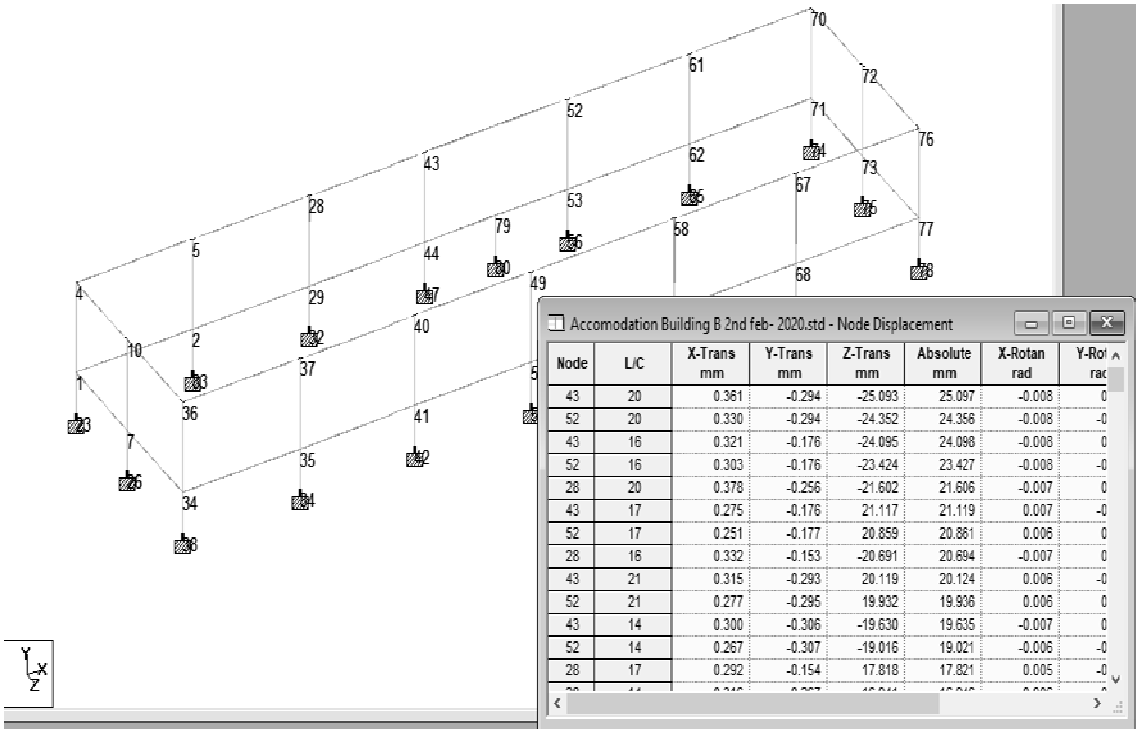
7. DEFLECTIONS/DRIFT

7.1. Maximum Allowable Deflection

L/240 : for member load with live load plus dead load

L/600 : for members supporting masonry under total loads

Where, L = span of member between support



Maximum attained Drift as per Analysis from above figure = 25mm

Allowable Drift as per Table-10.12, SBC-301-2007 is H/100 = 3300/100 = 33mm

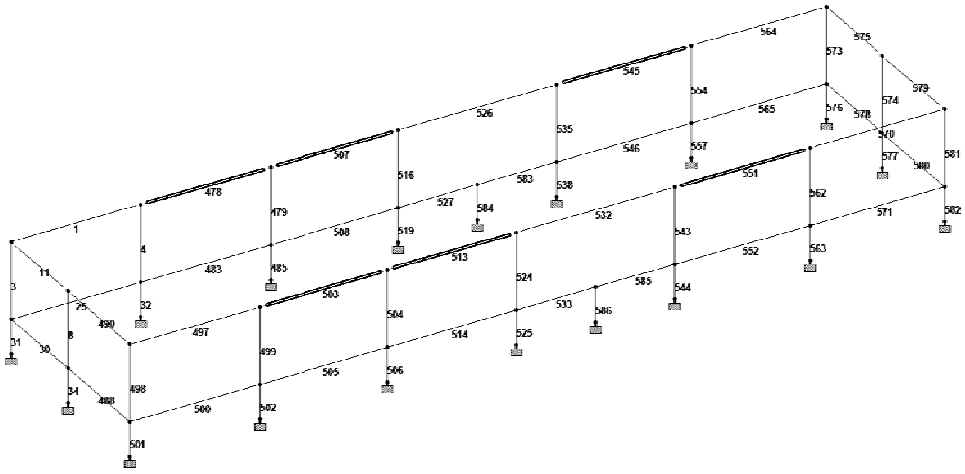
Hence the Structure is safe considering drift check…

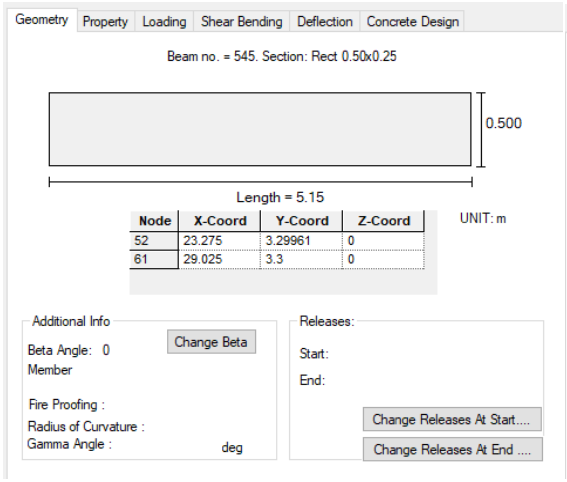
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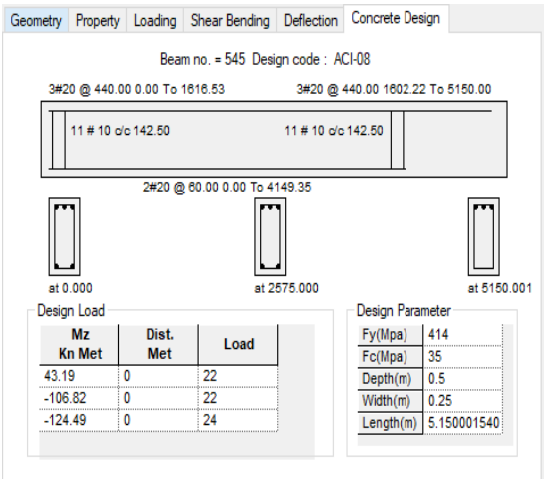
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8. CONCRETE ROOF BEAM DESIGN

8.1. Design of Ledger Beam(LB1)







Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = 124 **Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

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Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **800 mm^2**

Ast.min= (As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm^2**

Maximum Moment required from Analysis = **43 Knm (Sagging)**

Ast2.req = **266 mm^2**

Hence Provide 3-20 dia-Full Straight Bars Top & 3-20 dia-Full Straight Bottom

Ast1.pro= **942mm^2 >>>>>>(Ast.min, Ast1.req)**

Ast2.pro= **942mm^2 >>>>>>(Ast.min, Ast2.req)**

8.1.1. Staad Output For LB1

```
        B E A M   N O.    545 D E S I G N   R E S U L T S - SHEAR

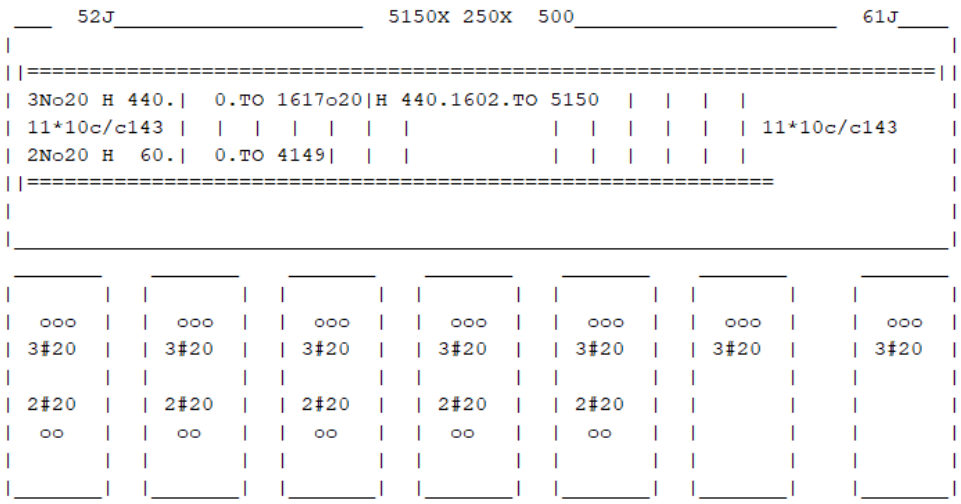

AT START SUPPORT - Vu=    54.47 KNS   Vc=  190.78 KNS   Vs=     0.00 KNS
Tu=    13.29 KN-MET   Tc=     3.9 KN-MET   Ts=    17.7 KN-MET   LOAD     26
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1395. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    4.38 SQ.CM.

AT END    SUPPORT - Vu=    57.50 KNS   Vc=  143.37 KNS   Vs=     0.00 KNS
Tu=    16.62 KN-MET   Tc=     3.8 KN-MET   Ts=    22.2 KN-MET   LOAD     27
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1395. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    5.47 SQ.CM.
```

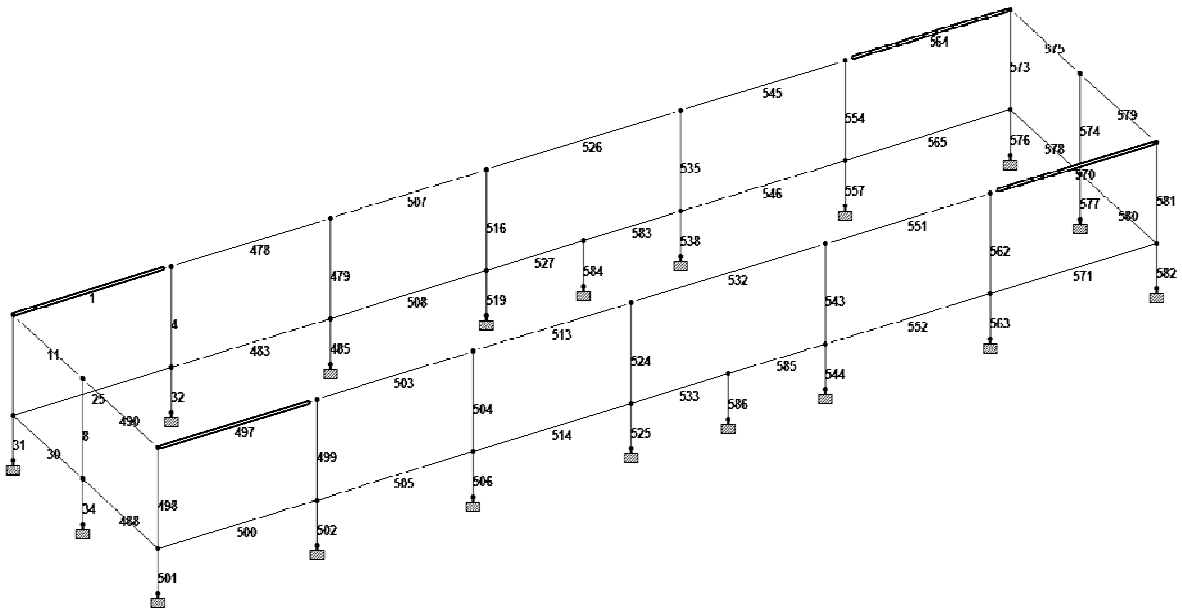

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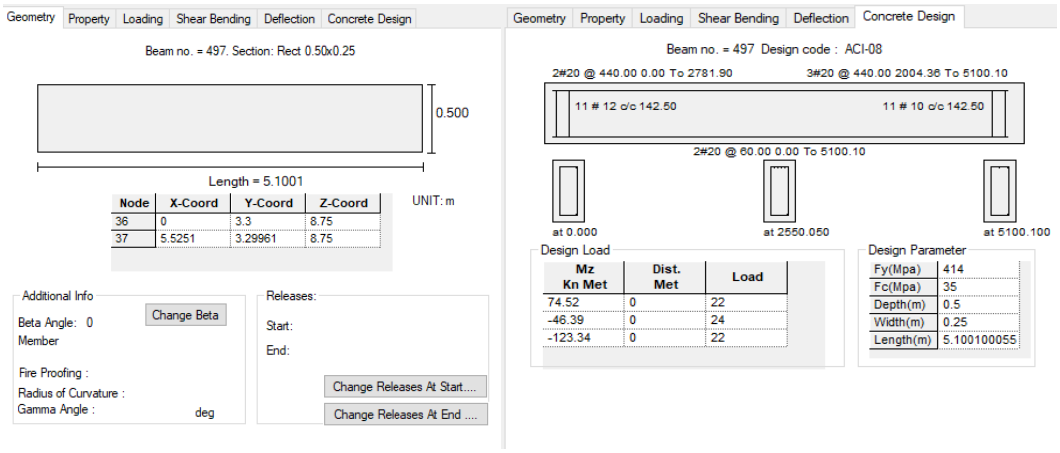


8.2. Design of Ledger Beam(LB2)



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Design of Ledger Beam (LB2)

Maximum Moment required from Analysis = 123 **Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **800 mm^2**

Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,min} = \frac{0.25\sqrt{f_c'}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm^2**

Maximum Moment required from Analysis = **74 Knm (Sagging)**

Ast2.req = **465 mm^2**

Hence Provide 3-20 dia-Full Straight Bars Top & 3-20 dia-Full Straight Bottom

Ast1.pro= **942mm^2 >>>>>>(Ast.min, Ast1.req)**

Ast2.pro= **942mm^2 >>>>>>(Ast.min, Ast2.req)**

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8.2.1. Staad Output For LB2

BEAM NO. 497 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

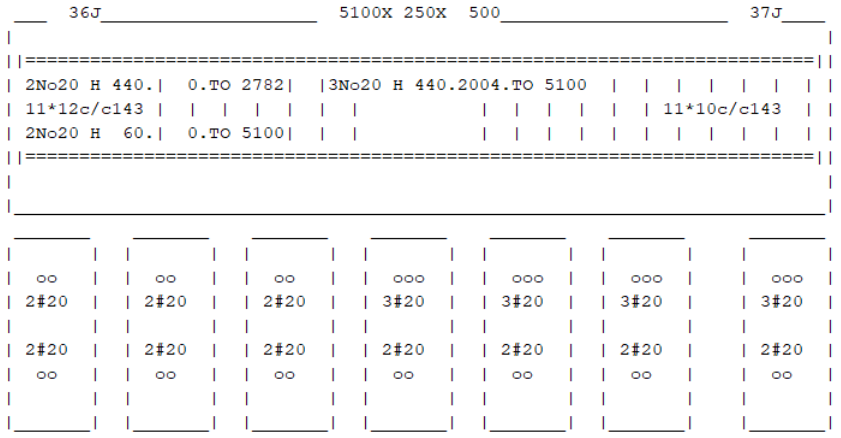
LEN - 5100. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	60.	2 - 20MM	0.	5100.	YES	YES
2	440.	2 - 20MM	0.	2782.	YES	NO
3	440.	3 - 20MM	2004.	5100.	NO	YES

B E A M N O. 497 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 38.62 KNS Vc= 105.94 KNS Vs= 0.00 KNS
Tu= 18.15 KN-MET Tc= 4.1 KN-MET Ts= 24.2 KN-MET LOAD 27
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 143. MM C/C FOR 1420. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 5.98 SQ.CM.

AT END SUPPORT - Vu= 72.91 KNS Vc= 195.58 KNS Vs= 0.00 KNS
Tu= 14.66 KN-MET Tc= 4.1 KN-MET Ts= 19.5 KN-MET LOAD 27
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM C/C FOR 1370. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 4.83 SQ.CM.

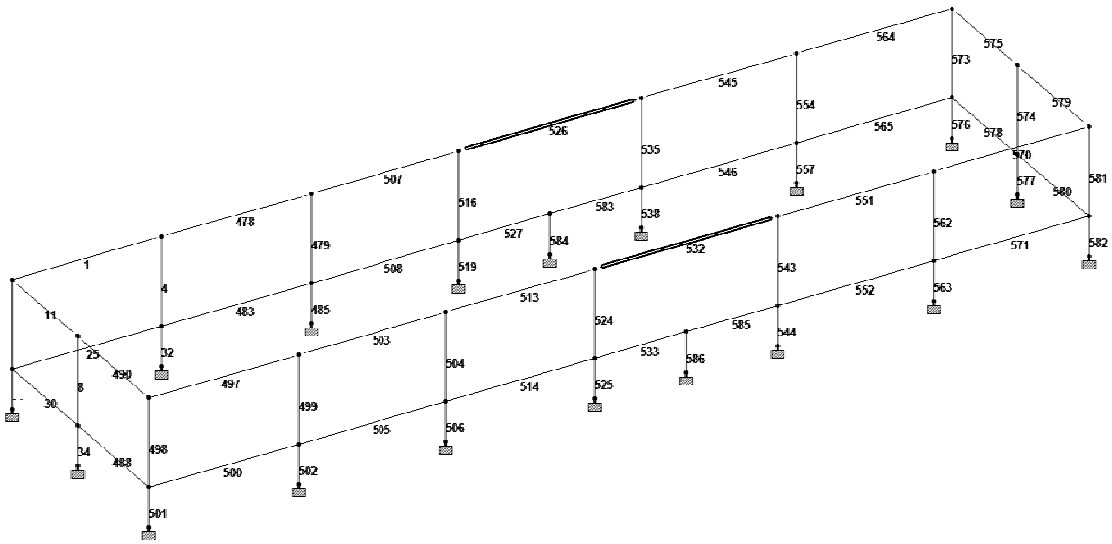


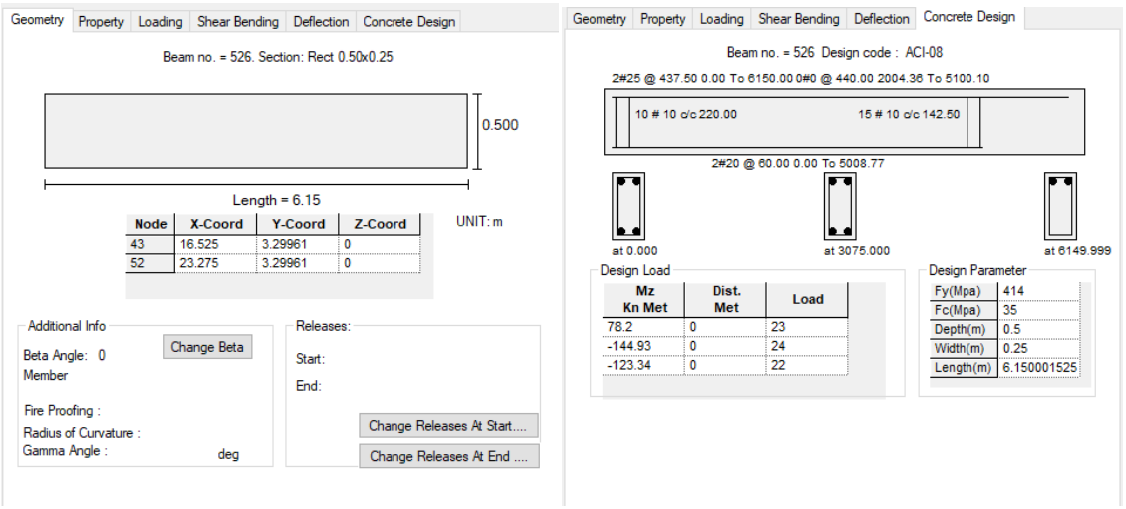
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8.3. Design of Ledger Beam(LB3)





Design of Ledger Beam (LB3)

Maximum Moment required from Analysis = 145 **Knm (Hogging)**

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Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **940 mm^2**

Ast.min= (As per clause-10.5-ACI-318)

$$A_{s,min} = \frac{0.25\sqrt{f_c'}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm^2**

Maximum Moment required from Analysis = **78 Knm (Sagging)**

Ast2.req = **470 mm^2**

Hence Provide 3-25 dia-Full Straight Bars Top & 3-20 dia-Full Straight Bottom

Ast1.pro= **1470mm^2 >>>>>>(Ast.min, Ast1.req)**

Ast2.pro= **942mm^2 >>>>>>(Ast.min, Ast2.req)**

8.3.1. Staad Output For LB3

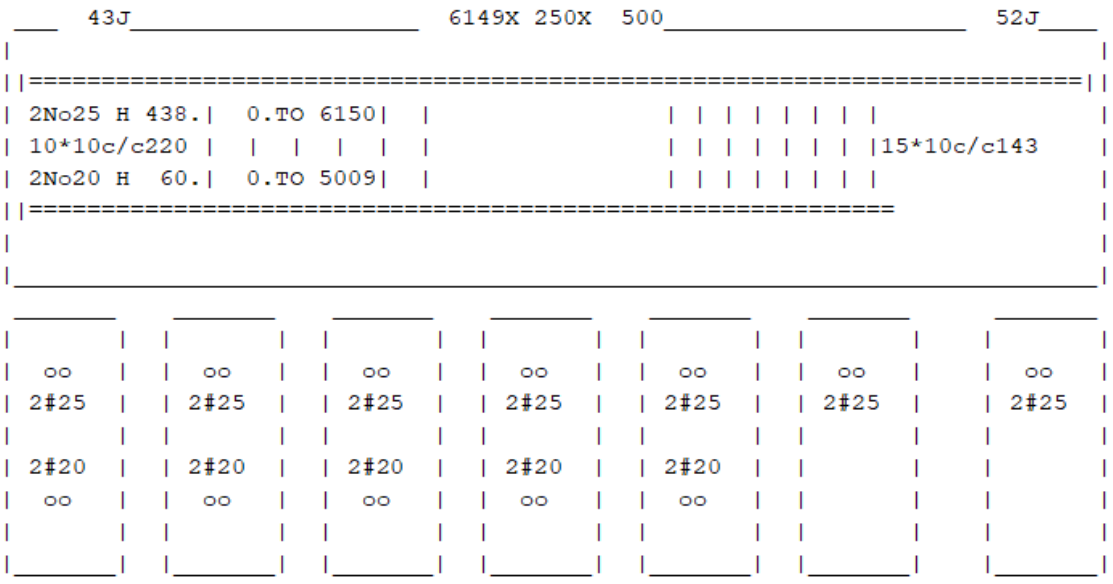
```
        B E A M   N O.    526 D E S I G N   R E S U L T S - SHEAR


AT START SUPPORT - Vu=    85.55 KNS   Vc=   194.86 KNS   Vs=     0.00 KNS
Tu=     2.38 KN-MET   Tc=     4.0 KN-MET   Ts=     0.0 KN-MET   LOAD     22
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 220. MM  C/C FOR 1895. MM

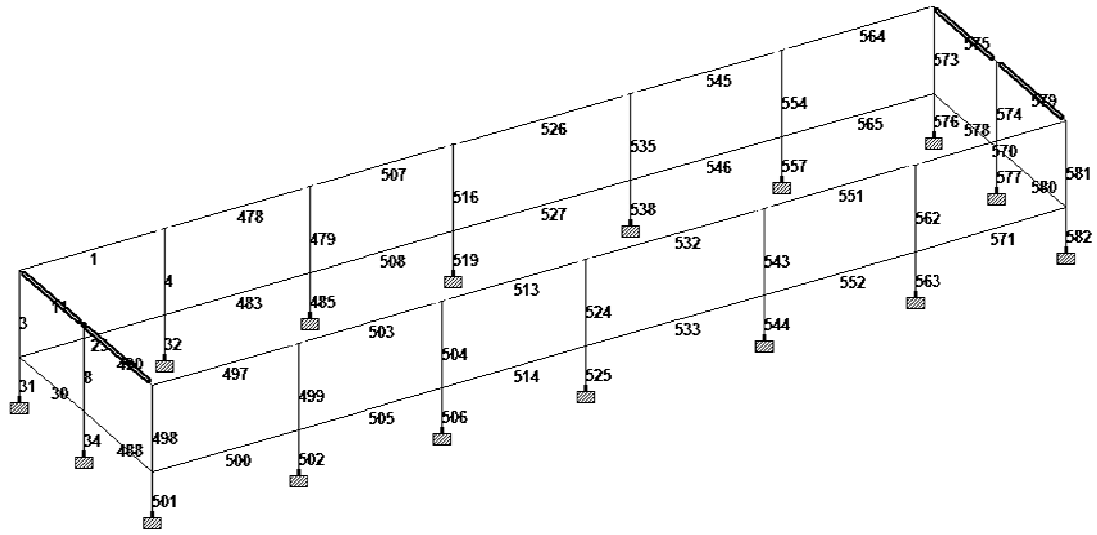
AT END    SUPPORT - Vu=    77.19 KNS   Vc=   193.74 KNS   Vs=     0.00 KNS
Tu=     4.32 KN-MET   Tc=     4.0 KN-MET   Ts=     5.8 KN-MET   LOAD     27
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1895. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    1.42 SQ.CM.
```


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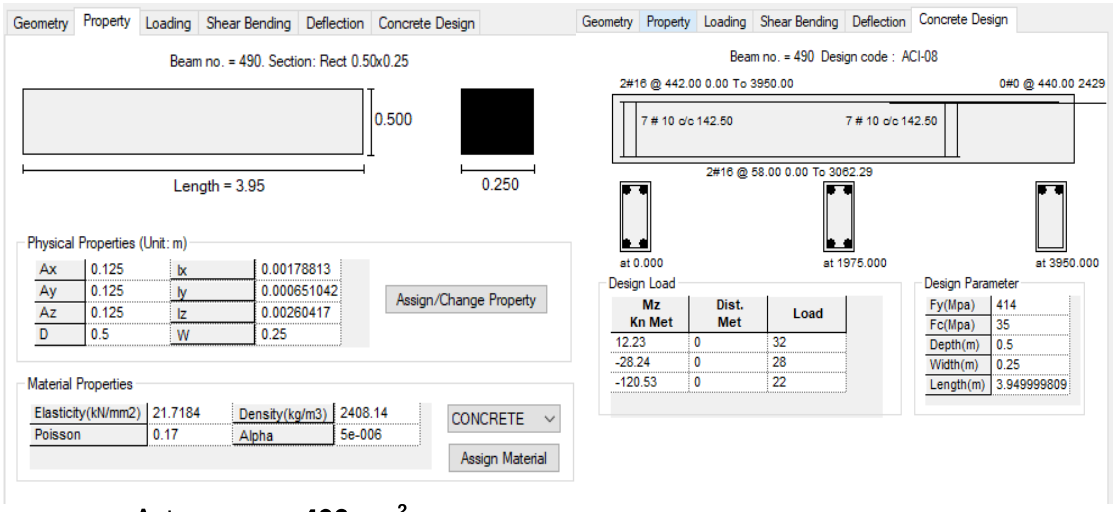
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8.4. Design of Beam-(B1)



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Ast.min= **406mm^2**

Hence Provide 3-16 dia-Full Straight Bars Top & Bottom

8.4.1. Staad Output For B1

```
        B E A M   N O.    490 D E S I G N   R E S U L T S - SHEAR

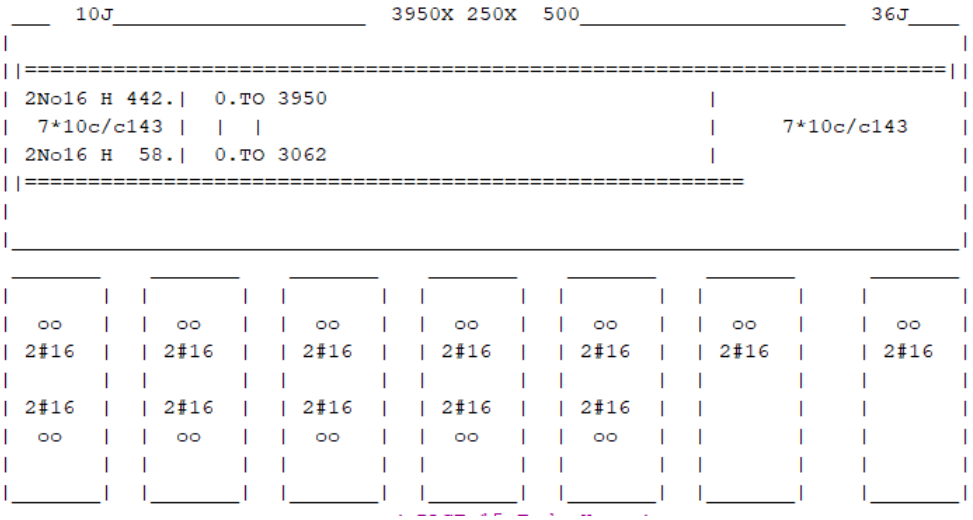

AT START SUPPORT - Vu=     1.97 KNS   Vc=   104.74 KNS   Vs=     0.00 KNS
Tu=     5.14 KN-MET   Tc=     3.7 KN-MET   Ts=     6.9 KN-MET   LOAD     31
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR  795. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    1.69 SQ.CM.

AT END    SUPPORT - Vu=     5.62 KNS   Vc=   105.63 KNS   Vs=     0.00 KNS
Tu=     4.85 KN-MET   Tc=     3.9 KN-MET   Ts=     6.5 KN-MET   LOAD     23
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR  795. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    1.60 SQ.CM.
```


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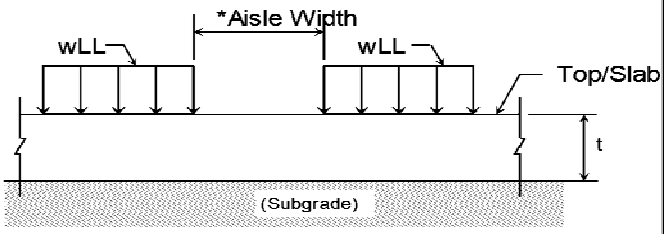
9. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm		
Concrete Strength, f 'c =	5000	psi	35.000	kpa		
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)	
Factor of Safety, FS =	1.500		1.500			
Uniform Live Load, wLL =	104.45	psf	5.00	kpa		



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Low. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) = 682.05 psf = 32.65 kpa wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)

wLL(allow) >= wLL, O.K.

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

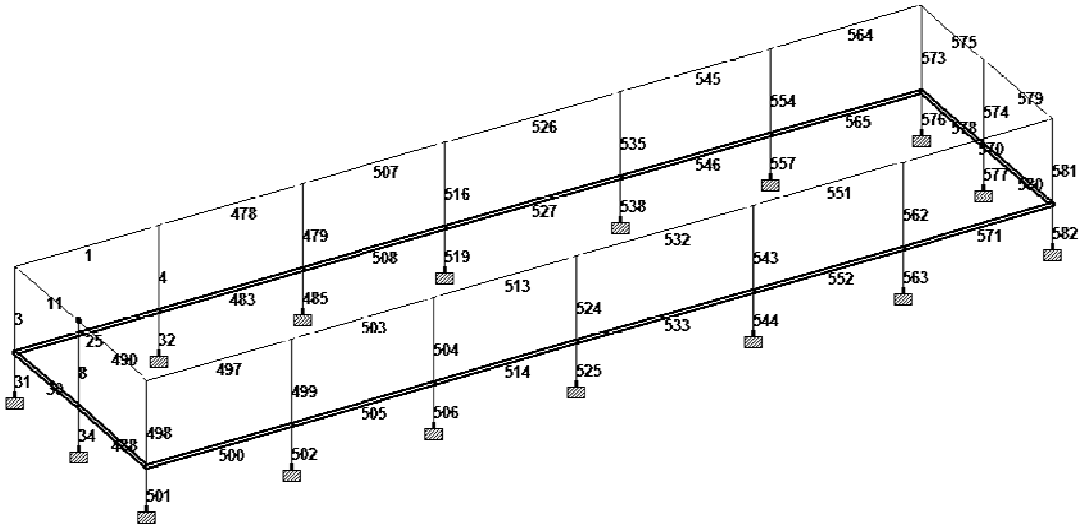
Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60{,}000}{f_y}$
			0.0014

Provide WWF-8mm Dia@200 c-c Both Ways-TOP LAYER

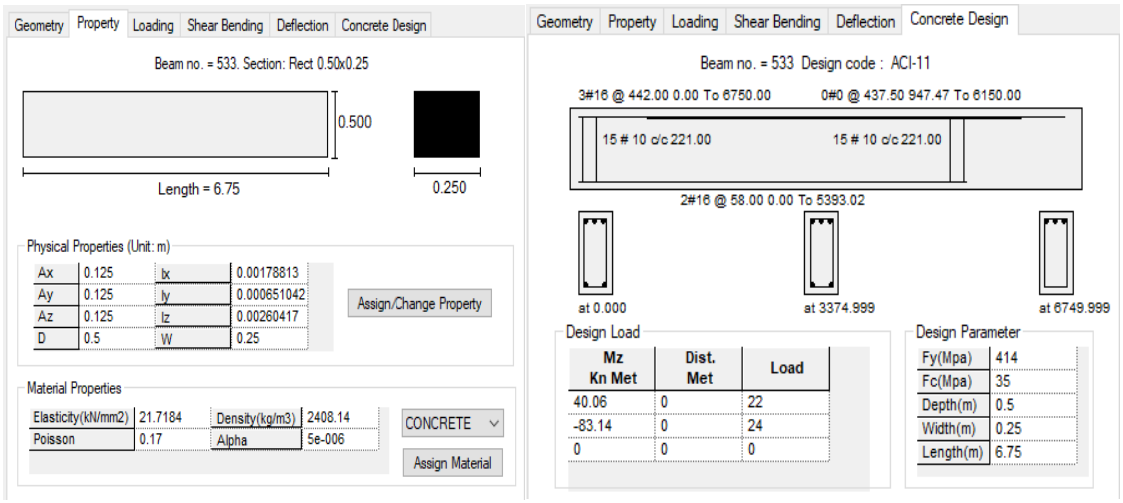
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10. DESIGN OF BEAM-(GB1)



MOMENT ENVELOPE FOR BEAMS GB1



Hence Provide 3-16 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bars Bottom

Ast.pro= **600mm^2 >>>>>>(Ast.min, Ast.req)**

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STAAD OUTPUT FOR GB1

BEAM NO. 533 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

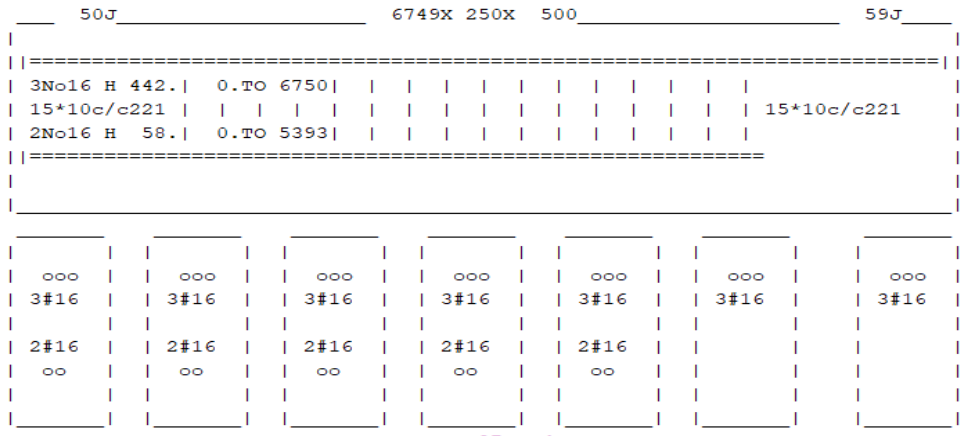
LEN - 6750. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	5393.	YES	NO
2	442.	3 - 16MM	0.	6750.	YES	YES

B E A M N O. 533 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 43.83 KNS Vc= 108.31 KNS Vs= 0.00 KNS
Tu= 0.05 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 2935. MM

AT END SUPPORT - Vu= 43.93 KNS Vc= 108.31 KNS Vs= 0.00 KNS
Tu= 0.05 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 2935. MM

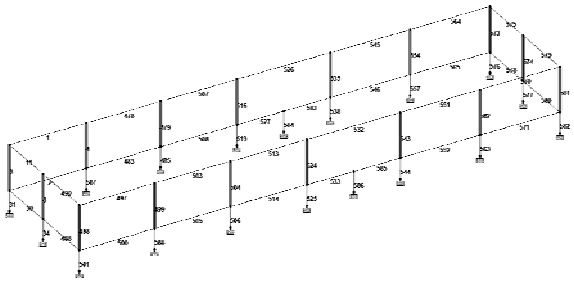


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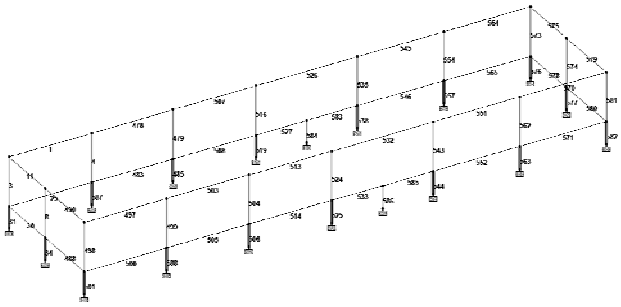
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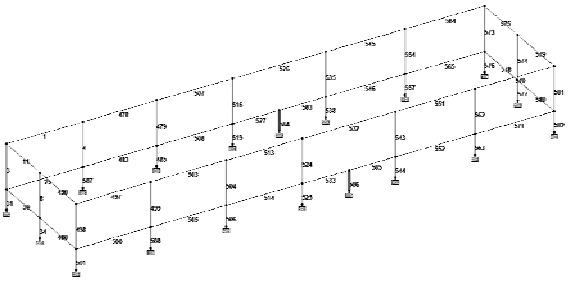
11. CONCRETE COLUMN DESIGN



Column numbers C1



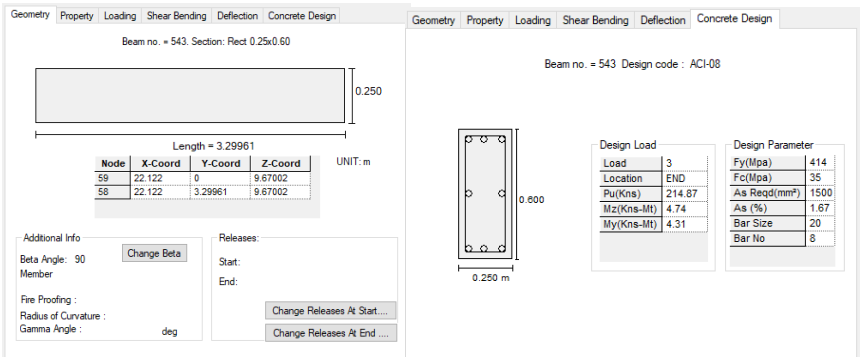
Column numbers CR



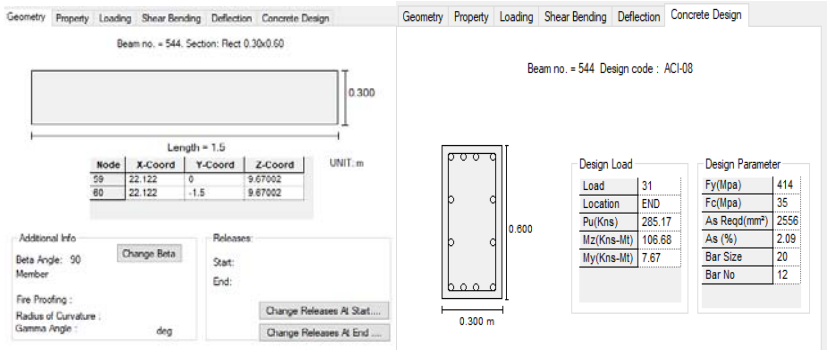
Column numbers CR1

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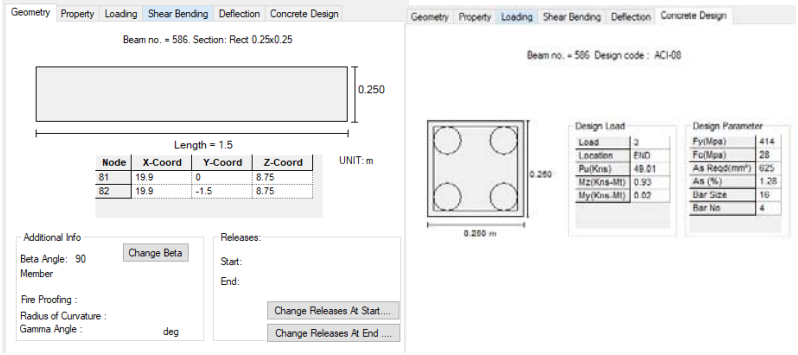
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Column Design-C1



Column Design-CR (Column Neck)



Column Design-CR1 (Column Neck)

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11.1. Staad Output For Column C1, CR & CR1

```
      COLUMN  NO.   562  DESIGN PER ACI 318-08 - AXIAL + BENDING


 FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 600.0 X 250.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  1500.0  SQ. MM

 BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ------------------------------------------------------------

   8 - 20 MM               1.676        3      END     0.650
 (PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
 TIE BAR NUMBER   10 SPACING 250.00 MM


 ===========================================================================
 ===========================================================================

      COLUMN  NO.   485  DESIGN PER ACI 318-08 - AXIAL + BENDING


 FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 600.0 X 300.0 MMS, TIED
        AREA OF STEEL REQUIRED =  3186.0  SQ. MM

 BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ------------------------------------------------------------

  12 - 20 MM               2.094       31      END     0.650
 (PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
 TIE BAR NUMBER   10 SPACING 300.00 MM


 ===========================================================================

      COLUMN  NO.   586  DESIGN PER ACI 318-08 - AXIAL + BENDING


 FY - 413.7  FC -  27.6 MPA,  SQRE SIZE - 250.0 X 250.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =   625.0  SQ. MM
-------------------------------< PAGE 52 Ends Here >----------------------------
  STAAD SPACE                                                -- PAGE NO.   53


 BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ------------------------------------------------------------

   4 - 16 MM               1.287        3      END     0.650
 (PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
 TIE BAR NUMBER   10 SPACING 250.00 MM
```

Hence Provide 8-20 dia-Vertical Reinf for Column Reinforcement-C1 &
Provide 8-25 dia-Vertical Reinf for Column Reinforcement-CR(Column Neck)
Provide 4-16 dia-Vertical Reinf for Column Reinforcement-CR1(Column Neck)

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12. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties			
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r	24		kN/m^3
Clear Span	0.6	met	Strength of concrete "f'_c"	35		N/mm^2
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420		N/mm^2
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20		mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9		
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75		
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es"	2E+05		N/mm^2
Loading			Modulus of Elasticity of concrete	27806		N/mm^2
Self Weight of slab	3.6	kN/m^2	Modular Ratio "n = Es/Ec"	7.19		
Floor Finish	3.5	kN/m^2	**Provided Reinforcement**			
Wall load	0	kN/m^2	Top layer Nos.	6	Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m^2	Bottom layer Nos.	0	Φ 12	mm
Live load	1	kN/m^2	Area of Top steel "$A_{s\ t}$"		468	mm^2
Total Ultimate Load "Uu_{tot}"$_{(1.2D+1.6L)}$	10.12	kN/m^2	Area of Bottom steel "$A_{s\ b}$"		0	mm^2
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"db" = 10 mm			
Calculations						
Shear force Vu = U*le	**8.096**	**kN**				
Neg. Moment Mu = U*le^2/2	**3.51**	**kNm**				

Check for Section capacity

Depth of comp. block "a" = $A_{s\ t}$*f_y/(0.85*f'_c*b)	6.61	mm					
Section capacity ΦMn = Φ*$A_{s\ t}$*f_y*(d-a/2)	**21.94**	**kNm**	**O.K**	**>**	**3.51**	**kNm**	(Mu)

Shear requirements

Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	**>**	**8.096**	**kN**	(Vu)

Serviceability requirements

d' = c+d_b/2	26	mm
B = b/(n*$A_{s\ t}$)	0.297	
r = (n-1)*$A_{s\ b}$/(n*$A_{s\ t}$)	0	
k_d = (√(2*d*B(1+r*d'/d)+(1+r)^2)-(1+r))/B	29.01	mm
I_{cr} = bk$_d$d^3/3+n*$A_{s\ t}$*(d-k_d)^2+(n-1)*$A_{s\ b}$*(k_d-d'^{2}	3.915E+07	mm^4
Calculate Gross moment of inertia		
Area of slab A = t*b	2E+05	mm^2
Depth of neutral axis "y"	75	mm
Distance of top fiber from N.A "y_t"	75	mm
Gross moment of inertia "I_g"	2.813E+08	mm^4
Allowable stress fr = 0.62√f'c	3.67	mm^2
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm

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Service moments

Moment due to dead load "M_{dl}"	2.54	kNm
Moment due to live load "Mll"	0.32	kNm
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm

Deflection due to dead + live loads

R(dl+ll) = M_{cr}/Ms	1.000		
$I_e(dl+ll) = R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
$\delta = K*5/48*Ms*le^2/(Ec*I_e)$	0.08	mm	**K = 3.2 (conservative)**

Deflection due to dead load

R(dl) = M_{cr}/Md	5.411	
$I_e(dl) = R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4
$\delta = K*5/48*Md*le^2/(Ec*I_e)$	0.00	mm

Deflection due to live load

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load =
per ACI 318 Tab 9.5b **$l_e/360$ 2.22 mm > 0.08 mm O.K**

Long term deflection

% of liveload sustained "SL"	50	%
$M_{sus} = M_{dl}+SL*M_{ll}$	2.70	kNm
R(sus) = $M_{cr}/Msus$	5.091	
$I_e(sus) = R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10	
$\delta = K*5/48*Msus*le^2/(Ec*I_e)$	0.00	mm

Magnification factor for creep and shrinkage

$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000

Where:-

ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318

$\rho' = A_{s\ b}/(b*d)$ 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

Allowable deflection = $l_e/240$ 3.333 mm > 0.079 mm O.K

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13. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-RPT-1008 for Details)

TABLE 10: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Village Area	BH-21	Village	Isolated Square	1.20	1.0 x 1.0	1000	350	350	25	42,000	32,200
				1.50		1100	500	500	25	60,000	38,900
				1.20	2.0 x 2.0	700	580	580	25	69,600	37,500
				1.50		850	750	750	25	90,000	44,200
				1.20	3.0 x 3.0	600	750	600	25	72,000	42,800
				1.50		700	930	700	25	84,000	49,500
				1.20	4.0 x 4.0	550	910	550	25	66,000	48,100
				1.50		625	1090	625	25	75,000	54,800
				1.20	5.0 x 5.0	475	1060	475	25	57,000	53,400
				1.50		550	1240	550	25	66,000	60,200

Refer Attachment- II for Foundation Design calculations

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REFERENCES

BUILDING PLANS	MD-216-6000-EG-AR-GAD-3087
OVERALL LAYOUT PLAN	MD-216-6000-EG-GAD-1535
LAYOUT & DETAILS OF FOUNDATION, COLUMN & GRADE BEAM	MD-216-6000-EG-CV-DED-1478
LAYOUT & DETAILS OF ROOF & GRADE SLAB	MD-216-6000-EG-CV-DED-1479
GEOTECHNICAL REPORT	MD-216-3000-EG-CV-RPT-1008
STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001

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ATTACHMENT 1 STAAD INPUT DATA

STAAD SPACE

START JOB INFORMATION

ENGINEER DATE 18-Oct-19

END JOB INFORMATION

INPUT WIDTH 79

UNIT METER KN

JOINT COORDINATES

1 0 0 0; 2 5.525 0 0; 4 0 3.3 0; 5 5.525 3.3 0; 7 0 0 4.2; 10 0 3.3 4.2;

23 0 -1.5 0; 26 0 -1.5 4.2; 28 11.025 3.29961 0; 29 11.025 0 0;

32 11.025 -1.5 0; 34 0 0 8.75; 35 5.5251 0 8.75; 36 0 3.3 8.75;

37 5.5251 3.29961 8.75; 38 0 -1.5 8.75; 40 11.025 3.3 8.75; 41 11.025 0 8.75;

42 11.025 -1.5 8.75; 43 16.525 3.29961 0; 44 16.525 0 0; 47 16.525 -1.5 0;

49 16.525 3.29961 8.75; 50 16.525 0 8.75; 51 16.525 -1.5 8.75;

52 23.275 3.29961 0; 53 23.275 0 0; 56 23.275 -1.5 0; 58 23.275 3.3 8.75;

59 23.275 0 8.75; 60 23.275 -1.5 8.75; 61 29.025 3.3 0; 62 29.025 0 0;

65 29.025 -1.5 0; 67 29.025 3.29961 8.75; 68 29.025 0 8.75;

69 29.025 -1.5 8.75; 70 34.775 3.29961 0; 71 34.775 0 0; 72 34.775 3.29961 4.2;

73 34.775 0 4.2; 74 34.775 -1.5 0; 75 34.775 -1.5 4.2; 76 34.775 3.29961 8.75;

77 34.775 0 8.75; 78 34.775 -1.5 8.75; 79 19.9 0 0; 80 19.9 -1.5 0;

81 19.9 0 8.75; 82 19.9 -1.5 8.75; 83 5.525 -1.5 0; 84 5.525 -1.5 8.75;

MEMBER INCIDENCES

1 4 5; 3 1 4; 4 2 5; 8 7 10; 11 4 10; 25 1 2; 30 7 1; 31 1 23; 34 7 26;

478 5 28; 479 29 28; 483 2 29; 485 29 32; 488 7 34; 490 10 36; 497 36 37;

498 34 36; 499 35 37; 500 35 34; 501 34 38; 503 37 40; 504 41 40; 505 41 35;

506 41 42; 507 28 43; 508 29 44; 513 40 49; 514 41 50; 516 44 43; 519 44 47;

524 50 49; 525 50 51; 526 43 52; 527 44 79; 532 49 58; 533 50 81; 535 53 52;

538 53 56; 543 59 58; 544 59 60; 545 52 61; 546 53 62; 551 58 67; 552 59 68;

554 62 61; 557 62 65; 562 68 67; 563 68 69; 564 61 70; 565 62 71; 570 67 76;

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571 68 77; 573 71 70; 574 73 72; 575 70 72; 576 71 74; 577 73 75; 578 71 73;

579 72 76; 580 73 77; 581 77 76; 582 77 78; 583 79 53; 584 79 80; 585 81 59;

586 81 82; 587 2 83; 588 35 84;

DEFINE MATERIAL START

ISOTROPIC CONCRETE

E 2.17184e+007

POISSON 0.17

DENSITY 23.6158

ALPHA 5e-006

DAMP 0.05

TYPE CONCRETE

STRENGTH FCU 27578.9

END DEFINE MATERIAL

MEMBER PROPERTY AMERICAN

31 34 485 501 506 519 525 538 544 557 563 576 577 582 587 -

588 PRIS YD 0.3 ZD 0.6

1 478 497 503 507 513 526 532 545 551 564 570 PRIS YD 0.5 ZD 0.25

MEMBER PROPERTY AMERICAN

25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 580 583 -

585 PRIS YD 0.5 ZD 0.25

MEMBER PROPERTY AMERICAN

11 490 575 579 PRIS YD 0.5 ZD 0.25

MEMBER PROPERTY AMERICAN

3 4 8 479 498 499 504 516 524 535 543 554 562 573 574 581 PRIS YD 0.25 ZD 0.6

MEMBER PROPERTY

584 586 PRIS YD 0.25 ZD 0.25

CONSTANTS

BETA 90 MEMB 4 479 485 499 504 506 516 519 524 525 535 538 543 544 554 557 -

562 563 584 586 TO 588

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MATERIAL CONCRETE MEMB 1 3 4 8 11 25 30 31 34 478 479 483 485 488 490 497 -

498 TO 501 503 TO 508 513 514 516 519 524 TO 527 532 533 535 538 543 TO 546 -

551 552 554 557 562 TO 565 570 571 573 TO 588

SUPPORTS

23 26 32 38 42 47 51 56 60 65 69 74 75 78 80 82 TO 84 FIXED

****

MEMBER CRACKED

3 4 8 31 34 REDUCTION RIY 0.700000 RIZ 0.700000

1 11 25 30 REDUCTION RIY 0.350000 RIZ 0.350000

MEMBER OFFSET

478 503 507 513 526 532 545 551 564 570 START 0.3 0 0

1 478 497 503 507 513 526 532 545 551 END -0.3 0 0

11 490 575 579 START 0 0 0.3

11 490 575 579 END 0 0 -0.3

1 497 START 0.125 0 0

564 570 END -0.125 0 0

DEFINE IBC 2003

SDS 0.1 SD1 0.03 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2 CT 0.04

SELFWEIGHT 1

MEMBER WEIGHT

25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 580 583 585 UNI 12

1 478 497 503 507 513 526 532 545 551 564 570 UNI 35

DEFINE WIND LOAD

TYPE 1 WIND X WINDWARD

<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!

ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -

1 3.500 M 33.000 M 10.000 M 2.000 0.010 0 -

0 0 1 1 1.030 1.000 1.000 0.850 0 -

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1 0 1 0.850 0.800 -0.550

!> END GENERATED DATA BLOCK

INT 1.59699 1.59699 1.59699 HEIG 0 4.572 4.572

EXP 0.8 JOINT 1 2 4 5 7 10 23 26 28 29 32 38 42 47 51 56 60 65 69 74 75 78 -

80 82 TO 84

TYPE 2 WIND X LEEWARD

<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!

ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -

1 3.500 M 35.000 M 10.000 M 2.000 0.010 1 -

0 0 1 1 1.030 1.000 1.000 0.850 0 -

1 0 1 0.850 -0.200 0.550

!> END GENERATED DATA BLOCK

INT -0.934826 -0.934826 -0.934826 HEIG 0 3.5 3.5

EXP 0.8 JOINT 1 2 4 5 7 10 23 26 28 29 32 38 42 47 51 56 60 65 69 74 75 78 -

80 82 TO 84

TYPE 3 WIND Z WINDWARD

<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!

ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -

1 4.000 M 35.000 M 10.000 M 2.000 0.010 0 -

0 0 1 1 1.030 1.000 1.000 0.850 0 -

1 1 1 0.850 0.800 -0.550

!> END GENERATED DATA BLOCK

INT 1.59699 1.59699 1.59699 HEIG 0 4.572 4.572

EXP 0.8 JOINT 1 2 4 5 7 10 23 26 28 29 32 38 42 47 51 56 60 65 69 74 75 78 -

80 82 TO 84

TYPE 4 WIND Z LEEWARD

<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!

ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -

1 4.000 M 35.000 M 10.000 M 2.000 0.010 1 -

0 0 1 1 1.030 1.000 1.000 0.850 0 -

1 0 1 0.850 -0.200 0.550

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```
!> END GENERATED DATA BLOCK
INT -0.934826 -0.934826 -1.26591 HEIG  0  4  4
EXP 0.8 JOINT 1 2 4 5 7 10 23 26 28 29 32 38 42 47 51 56 60 65 69 74 75 78 -
80 82 TO 84
LOAD 1 LOADTYPE Seismic  TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic  TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 580 583 -
585 UNI GY -12
1 478 497 503 507 513 526 532 545 551 564 570 UNI GY -29.45
11 490 UMOM GZ 1
575 579 UMOM GZ -1
1 478 507 526 545 564 UMOM GX -1
497 503 513 532 551 570 UMOM GX 1
LOAD 4 LOADTYPE Roof Live  TITLE LL
MEMBER LOAD
1 478 497 503 507 513 526 532 545 551 564 570 UNI GY -2.1875
11 490 UMOM GZ 0.1
575 579 UMOM GZ -0.1
1 478 507 526 545 564 UMOM GX -0.1
497 503 513 532 551 570 UMOM GX 0.1
LOAD 5 LOADTYPE Wind  TITLE WL+X
WIND LOAD X 1 TYPE 1 XR 0 0.2 YR 0 4 ZR 0 10
```


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WIND LOAD -X 1 TYPE 2 XR 34.776 35 YR 0 4

WIND LOAD Z -1 TYPE 3 XR 0 35 YR 0 4 ZR 0 0.1

WIND LOAD -Z 1 TYPE 4 YR 0 4 ZR 8.74 9

LOAD 6 LOADTYPE Wind TITLE WL+Z

WIND LOAD Z 1 TYPE 3 XR -0.1 35 YR 0 4 ZR 0 0.1

WIND LOAD -Z 1 TYPE 4 YR 0 4 ZR 8.74 9

WIND LOAD X 1 TYPE 1 XR -0.1 0.122 YR 0 4 ZR -0.6 10

WIND LOAD -X 1 TYPE 2 XR 34.776 35 YR 0 4 ZR -0.6 11

LOAD 7 LOADTYPE Temperature TITLE TEMP

TEMPERATURE LOAD

1 3 4 8 11 478 479 490 497 TO 499 503 504 507 513 516 524 526 532 535 543
-

545 551 554 562 564 570 573 TO 575 579 581 TEMP 25

LOAD COMB 8 SLS: (DL)

3 1.0

LOAD COMB 9 SLS: (DL+T)

3 1.0 7 1.0

LOAD COMB 10 SLS: (D+LR)

3 1.0 4 1.0

LOAD COMB 11 SLS: (D+0.75LR)

3 1.0 4 0.75

LOAD COMB 12 SLS: (D+0.7EZ)

3 1.0 1 0.7

LOAD COMB 13 SLS: (D+0.7EX)

3 1.0 2 0.7

LOAD COMB 14 SLS: (D+0.75WX+0.75LR)

3 1.0 5 0.75 4 0.75

LOAD COMB 15 SLS: (D+0.75WZ+0.75LR)

3 1.0 6 0.75 4 0.75

LOAD COMB 16 SLS: (0.6D+WX)

3 0.6 5 1.0

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LOAD COMB 17 SLS: (0.6D+WZ)

3 0.6 6 1.0

LOAD COMB 18 SLS: (0.6D+0.7EX)

3 0.6 2 0.7

LOAD COMB 19 SLS: (0.6D+0.7EZ)

3 0.6 1 0.7

LOAD COMB 20 SLS: (D+WX)

3 1.0 5 1.0

LOAD COMB 21 SLS: (D+WZ)

3 1.0 6 1.0

LOAD COMB 22 ULS: (1.4 DL)

3 1.4

LOAD COMB 23 ULS: (1.4D+1.4T+0.5LR)

3 1.4 4 0.5 7 1.4

LOAD COMB 24 ULS: (1.4D+1.4T+0.5LR)

3 1.4 4 0.5 7 1.4

LOAD COMB 25 ULS: (1.2D+0.80WX+1.6LR)

3 1.2 5 0.8 4 1.6

LOAD COMB 26 ULS: (1.2D+1.6WZ+0.5LR)

3 1.2 6 1.6 4 0.5

LOAD COMB 27 ULS: (1.2D+1.6WX+0.5LR)

3 1.2 4 0.5 5 1.6

LOAD COMB 28 ULS: (1.2D+1.6WZ+0.5LR)

3 1.2 4 0.5 6 1.6

LOAD COMB 29 ULS: (1.2D+EX)

3 1.2 2 1.0

LOAD COMB 30 ULS: (1.2D+EZ)

3 1.2 1 1.0

LOAD COMB 31 ULS: (0.90D+1.6WX)

3 0.9 5 1.6

LOAD COMB 32 ULS: (0.90D+1.6WZ)

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3 0.9 6 1.6

LOAD COMB 33 ULS: (0.90D+EZ)

3 0.9 1 1.0

LOAD COMB 34 ULS: (0.90D+EX)

3 0.9 2 1.0

PERFORM ANALYSIS

START CONCRETE DESIGN

CODE ACI

FC 35000 MEMB 1 3 4 8 11 25 30 31 34 478 479 483 485 488 490 497 TO 501 503 -

504 TO 508 513 514 516 519 524 TO 527 532 533 535 538 543 TO 546 551 552 554 -

557 562 TO 565 570 571 573 TO 582

CLB 0.04 MEMB 1 3 4 8 11 25 30 31 34 478 479 483 485 488 490 497 TO 501 503 -

504 TO 508 513 514 516 519 524 TO 527 532 533 535 538 543 TO 546 551 552 554 -

557 562 TO 565 570 571 573 TO 582

CLS 0.04 ALL

CLT 0.04 MEMB 1 3 4 8 11 25 30 31 34 478 479 483 485 488 490 497 TO 501 503 -

504 TO 508 513 514 516 519 524 TO 527 532 533 535 538 543 TO 546 551 552 554 -

557 562 TO 565 570 571 573 TO 582

MAXMAIN 20 MEMB 3 4 8 31 34 479 485 498 499 501 504 506 516 519 524 525 535 -

538 543 544 554 557 562 563 573 574 576 577 581 582 584 586 TO 588

MINMAIN 20 MEMB 3 4 8 31 34 479 485 498 499 501 504 506 516 519 524 525 535 -

538 543 544 554 557 562 563 573 574 576 577 581 582 584 586 TO 588

MAXMAIN 25 MEMB 1 11 478 490 497 503 507 513 526 532 545 551 564 570 575 579

MINMAIN 20 MEMB 1 11 478 490 497 503 507 513 526 532 545 551 564 570 575 579

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MAXMAIN 16 MEMB 11 490 575 579 584 586

MINMAIN 16 MEMB 11 490 575 579 584 586

MAXMAIN 16 MEMB 25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 -

580 583 585

MINMAIN 16 MEMB 25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 -

580 583 585

MINSEC 10 ALL

EFACE 0.74 MEMB 1 478 497 503 507 513 526 532 545 551

SFACE 0.74 MEMB 478 503 507 513 526 532 545 551 564 570

EFACE 0.74 MEMB 11 490 575 579

SFACE 0.74 MEMB 11 490 575 579

EFACE 0.69 MEMB 564 570

SFACE 0.69 MEMB 1 497

DESIGN BEAM 1 11 25 30 478 483 488 490 497 500 503 505 507 508 513 514 526 -

527 532 533 545 546 551 552 564 565 570 571 575 578 TO 580 583 585

DESIGN COLUMN 3 4 8 31 34 479 485 498 499 501 504 506 516 519 524 525 535 -

538 543 544 554 557 562 563 573 574 576 577 581 582 584 586 TO 588

END CONCRETE DESIGN

FINISH

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ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

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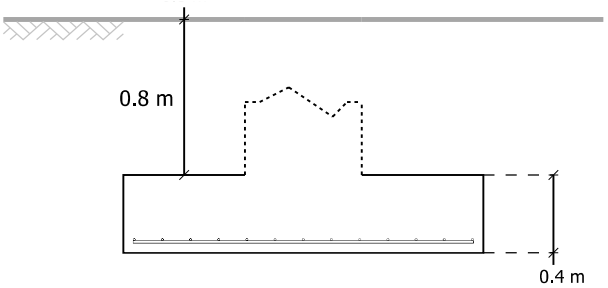
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Isolated Footing Design(ACI 318-08)

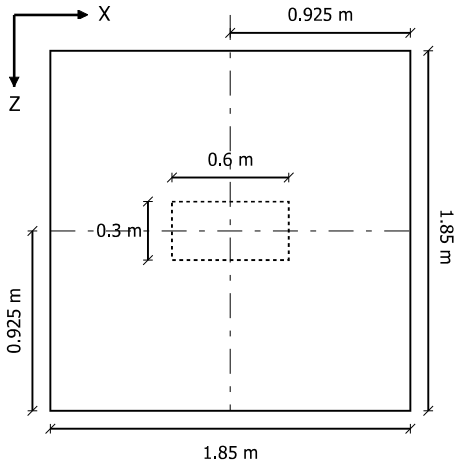
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
47	1	1.850m	1.850m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
47	13 - ϕ12	13 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F1-ALONG GRID 1 & D



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1850.000mm
Footing Width - Z (Fw) : 1850.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.600m

Column Width - Z (B_{col}) : 0.300m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 140.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.850m

Initial Width (W_o) = 1.850m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
8	SLS: (DL)	1.00	1.00	1.00
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
8	292.393	-4.467	0.009	-4.726	1.156
9	288.454	-5.230	0.018	-4.834	3.479
10	304.706	-4.223	0.010	-5.198	1.121
11	301.627	-4.284	0.010	-5.080	1.130
12	292.389	-4.525	5.176	16.430	1.255
13	294.250	5.607	0.010	-4.728	-15.588
14	302.110	-1.266	-23.522	-73.228	-3.525
15	302.051	-1.672	23.548	63.090	-2.840
16	176.079	1.344	-31.370	-93.699	-5.513
17	176.000	0.803	31.390	88.058	-4.600
18	177.293	7.394	0.006	-2.838	-16.051
19	175.432	-2.738	5.172	18.320	0.793
20	293.036	-0.442	-31.366	-95.589	-5.050
21	292.957	-0.984	31.393	86.168	-4.137

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	409.350	-6.254	0.013	-6.616	1.619
23	409.992	-7.201	0.025	-7.004	4.853
24	409.992	-7.201	0.025	-7.004	4.853

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
25	371.086	-1.751	-25.088	-79.118	-3.634
26	357.931	0.334	50.226	139.523	-7.100
27	358.057	1.201	-50.189	-151.289	-8.560
28	357.931	0.334	50.226	139.523	-7.100
29	353.524	9.032	0.012	-5.674	-22.534
30	350.866	-5.443	7.392	24.552	1.529
31	264.182	2.419	-50.193	-149.635	-8.890
32	264.057	1.552	50.223	141.177	-7.429
33	263.148	-4.103	7.389	25.970	1.182
34	265.806	10.372	0.009	-4.256	-22.880

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 3.422m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.989m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.850 m Governing Load Case : # 8

Width (W_2) = 1.850 m Governing Load Case : # 8

Depth (D_2) = 0.400 m Governing Load Case : # 27

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 3.422 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 32.855 kN

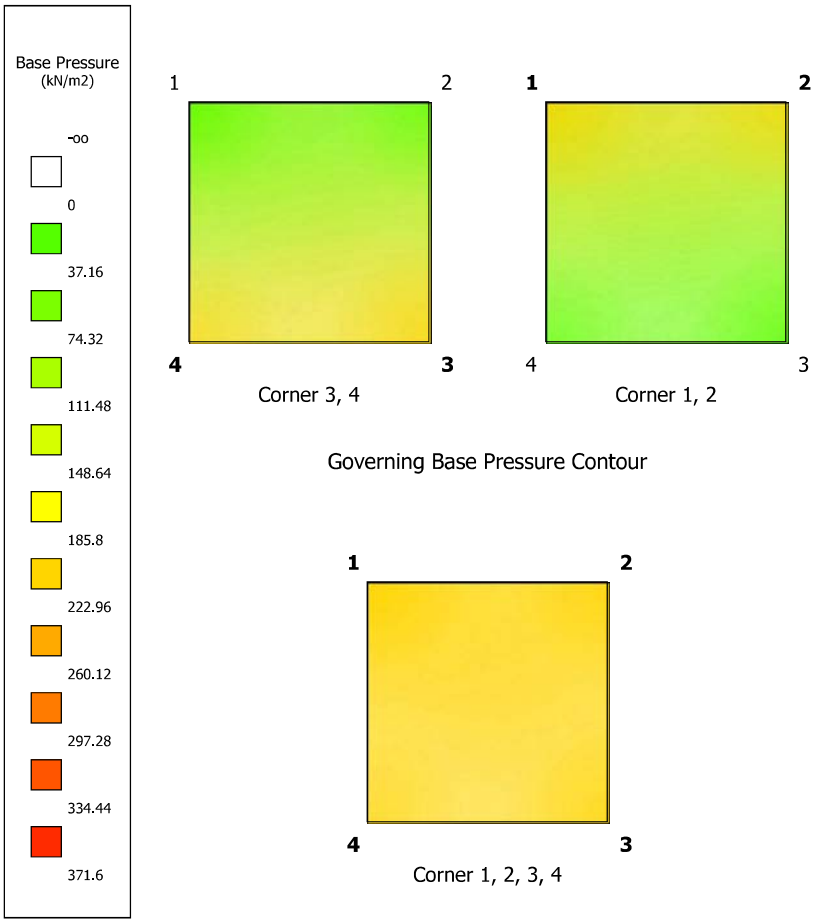
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 46.690 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



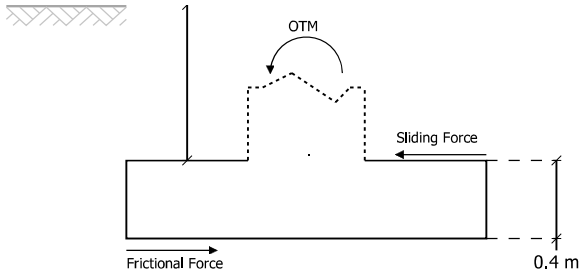
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
20	**206.7181**	215.9540	11.0063	1.7704	0.000
20	206.7181	**215.9540**	11.0063	1.7704	0.000
21	11.7356	18.8308	**205.9429**	198.8477	0.000
21	11.7356	18.8308	205.9429	**198.8477**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
20	**206.7181**	215.9540	11.0063	1.7704
20	206.7181	**215.9540**	11.0063	1.7704
21	11.7356	18.8308	**205.9429**	198.8477
21	11.7356	18.8308	205.9429	**198.8477**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
8	44.130	21006.721	62.409	1.500	73.237	116.894	1.500
9	37.289	11046.642	52.734	1.500	70.889	61.097	1.500
10	48.221	20179.646	68.194	1.500	68.771	126.474	1.500
11	47.155	20375.341	66.686	1.500	69.810	123.995	1.500
12	43.563	38.087	40.550	1.500	18.693	112.238	1.500
13	35.331	20555.229	49.965	1.500	73.568	19.390	1.500
14	159.805	8.600	12.144	1.500	4.294	116.946	1.500
15	120.960	8.589	12.116	1.500	4.893	162.541	1.500
16	100.771	4.319	6.102	1.500	2.242	39.080	1.500
17	168.709	4.315	6.100	1.500	2.367	48.035	1.500
18	18.410	23134.614	26.035	1.500	84.422	12.498	1.500
19	49.351	26.129	32.658	1.500	11.656	124.923	1.500
20	446.407	6.296	8.902	1.500	3.204	70.721	1.500
21	200.638	6.289	8.889	1.500	3.508	92.039	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 18
Governing Disturbing Force : 7.394kN
Governing Restoring Force : 136.124kN
Minimum Sliding Ratio for the Critical Load Case : 18.410
Critical Load Case for Overturning about X-Direction : 16
Governing Overturning Moment : -106.247kNm
Governing Resisting Moment : 238.246kNm
Minimum Overturning Ratio for the Critical Load Case : 2.242

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 17
Governing Disturbing Force : 31.390kN
Governing Restoring Force : 135.439kN
Minimum Sliding Ratio for the Critical Load Case : 4.315
Critical Load Case for Overturning about Z-Direction : 18
Governing Overturning Moment : -19.009kNm
Governing Resisting Moment : 237.571kNm
Minimum Overturning Ratio for the Critical Load Case : 12.498

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 17

Governing Disturbing Force : 31.400kN

Governing Restoring Force : 191.540kN

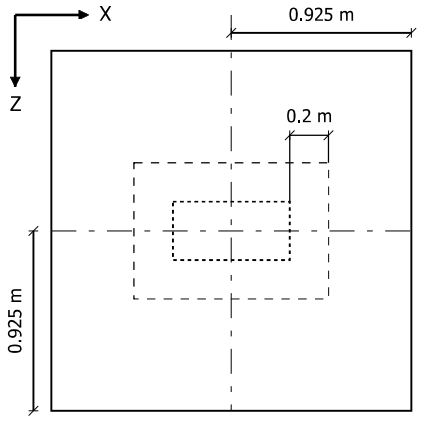
Minimum Sliding Ratio for the Critical Load Case : 6.100

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 415.393kN, Load Case # 27

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.176m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.805kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3398.657kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.805kN

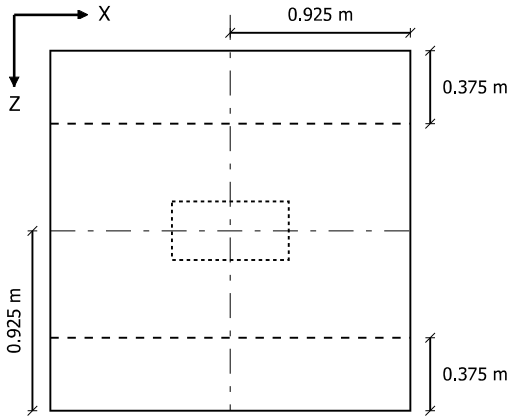
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1610.104kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 625.250kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.431m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

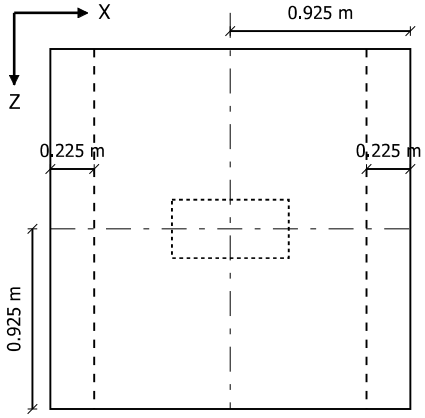
From above calculations, 0.75 X V_c = 468.938 kN

Critical load case for V_{ux} is # 27 $V_{ux} = V_{ux}|_{x=D_x}$ = 201.266 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 625.250 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.281 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

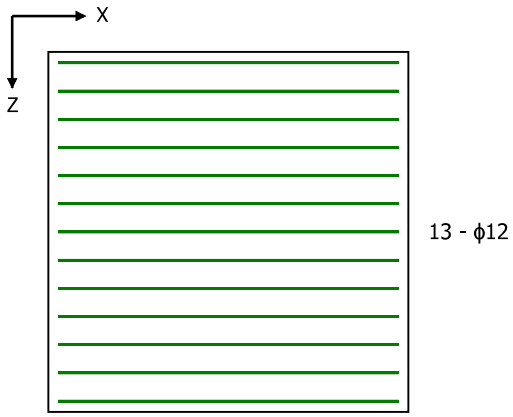
From above calculations, 0.75 X V_c = 468.938 kN

Critical load case for V_{uz} is # 23 $V_{uz} = V_{uz}|_{z=D_z}$ = 77.588 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 23

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00163
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.625 m
Ultimate moment,	$M_u\big\vert_{z=D_x}$ =	53.733 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	59.703 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00065
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00056
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1208.355 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 140.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}} =$

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover} =$ 0.575 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}} =$ 13

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1470.255 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)} =$ 0.00231

From ACI Cl.7.6.1, minimum req'd clear distance between bars

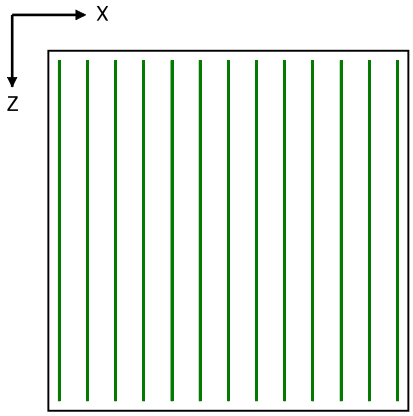
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 140.000mm

Provided Steel Area / Required Steel Area = 1.217

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



13 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 27

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal} =$ 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00170

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')} =$ 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.775 m

Ultimate moment, $M_u|_{x=D_x}$ = 136.038 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 151.153 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00179

(Based on gross depth) ρ x d_{eff} / Depth = 0.00148

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1257.042 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 140.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.725 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 13

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1470.255 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

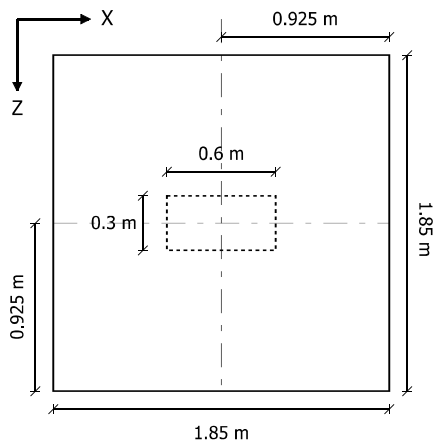
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00239

From ACI Cl.7.6.1, minimum req'd clear distance between bars

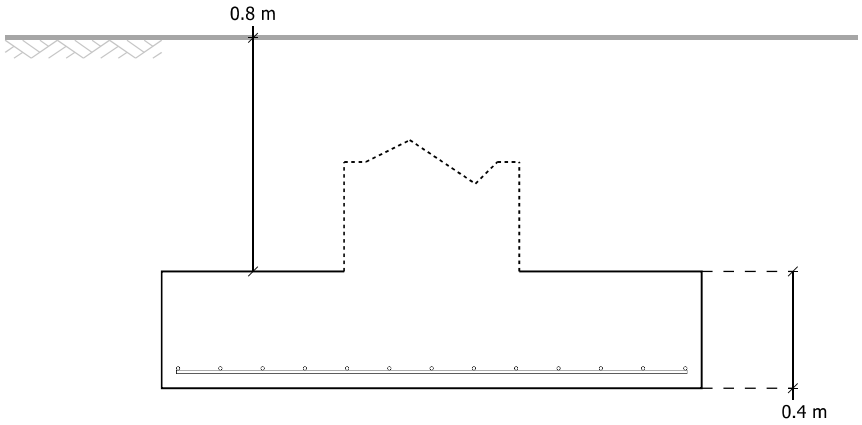
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 140.000mm

Provided Steel Area / Required Steel Area = 1.170

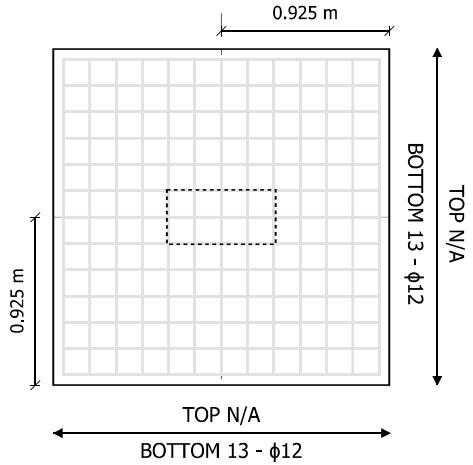
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.85 m	1.85 m	0.4 m	N/A	N/A	13 - ϕ12	13 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

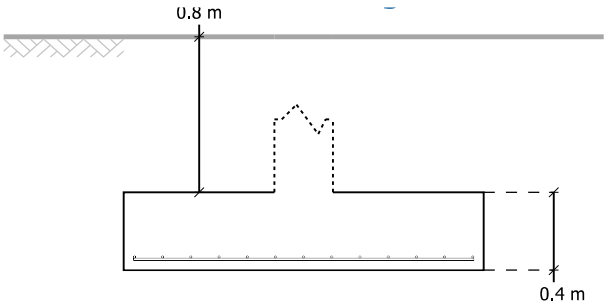
Print Calculation Sheet

Isolated Footing Design(ACI 318-08)

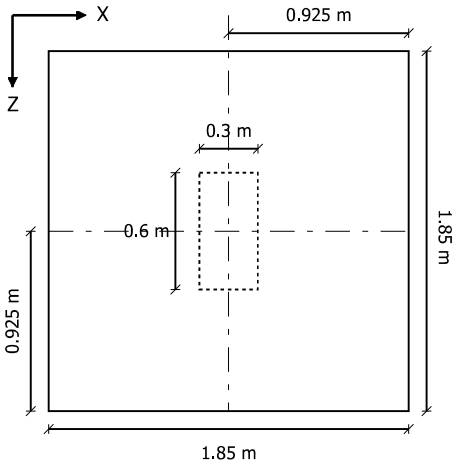
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
26	1	1.850m	1.850m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
26	13 - ϕ12	13 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F1-ALONG GRID 2 & A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1850.000mm
Footing Width - Z (Fw) : 1850.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.300m

Column Width - Z (B_{col}) : 0.600m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 140.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.850m

Initial Width (W_o) = 1.850m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
8	SLS: (DL)	1.00	1.00	1.00
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
8	96.497	-1.180	-1.469	-0.073	0.014
9	92.201	-2.130	-0.965	-0.308	3.562
10	96.474	-1.222	-1.455	-0.058	0.061
11	96.480	-1.212	-1.459	-0.062	0.049
12	94.347	-1.047	2.077	5.734	-0.299
13	96.462	0.283	-1.473	-0.087	-2.986
14	95.231	9.080	-2.677	-2.035	-16.714
15	92.790	7.288	6.189	11.562	-12.852
16	56.234	13.014	-2.506	-2.675	-22.342
17	52.978	10.624	9.315	15.455	-17.193
18	57.863	0.755	-0.885	-0.058	-2.991
19	55.748	-0.575	2.664	5.764	-0.305
20	94.833	12.542	-3.094	-2.705	-22.337
21	91.577	10.152	8.728	15.425	-17.188

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	135.096	-1.652	-2.056	-0.103	0.020
23	129.070	-3.003	-1.344	-0.424	5.010
24	129.070	-3.003	-1.344	-0.424	5.010

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
25	114.428	9.494	-3.041	-2.168	-17.790
26	107.913	16.695	14.559	24.718	-27.483
27	113.122	20.519	-4.356	-4.290	-35.721
28	107.913	16.695	14.559	24.718	-27.483
29	115.746	0.674	-1.768	-0.107	-4.268
30	112.724	-1.226	3.302	8.209	-0.431
31	84.184	20.894	-3.922	-4.276	-35.749
32	78.975	17.070	14.992	24.732	-27.510
33	83.775	-0.872	3.743	8.231	-0.435
34	86.797	1.028	-1.328	-0.085	-4.273

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 3.422m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 0.811m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.850 m Governing Load Case : # 8

Width (W_2) = 1.850 m Governing Load Case : # 8

Depth (D_2) = 0.400 m Governing Load Case : # 27

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 3.422 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 32.855 kN

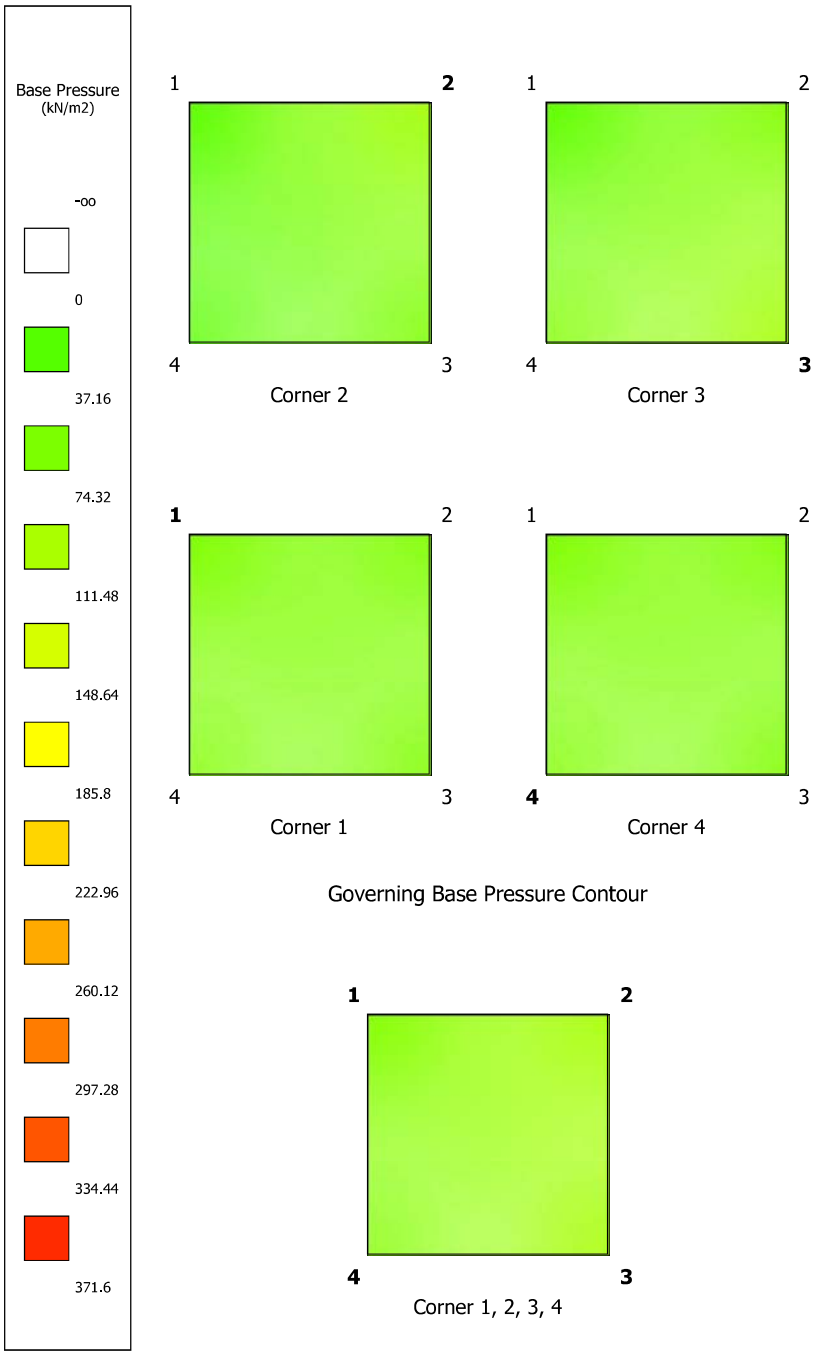
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 46.690 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



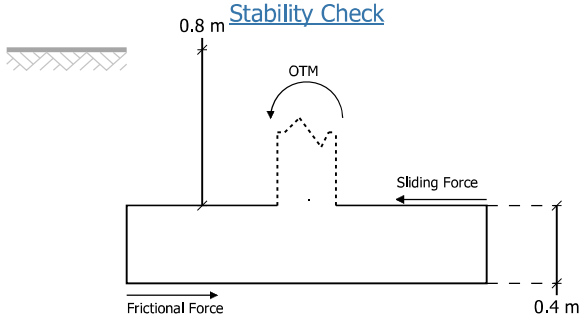
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
9	**55.0220**	46.6565	45.3409	53.7064	0.000
20	28.7649	**80.6076**	73.1359	21.2932	0.000
21	11.9374	52.2092	**88.0609**	47.7891	0.000
12	44.7003	44.4739	56.9166	**57.1429**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
9	**55.0220**	46.6565	45.3409	53.7064
20	28.7649	**80.6076**	73.1359	21.2932
21	11.9374	52.2092	**88.0609**	47.7891
12	44.7003	44.4739	56.9166	**57.1429**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
8	79.063	63.525	70.033	1.500	246.404	338.388	1.500
9	42.739	94.337	55.055	1.500	228.867	36.399	1.500
10	76.321	64.106	69.420	1.500	254.443	299.524	1.500
11	76.989	63.959	69.575	1.500	252.384	308.379	1.500
12	88.017	44.381	56.043	1.500	24.500	1361.839	1.500
13	329.452	63.332	87.955	1.500	240.803	53.118	1.500
14	10.202	34.597	13.838	1.500	52.046	8.034	1.500
15	12.533	14.759	13.510	1.500	11.356	10.224	1.500
16	5.529	28.712	7.679	1.500	34.148	4.624	1.500
17	6.611	7.540	7.030	1.500	6.391	5.800	1.500
18	96.434	82.251	88.501	1.500	308.686	39.138	1.500
19	124.693	26.915	37.207	1.500	18.324	1688.491	1.500
20	7.369	29.872	10.118	1.500	40.915	5.962	1.500
21	8.933	10.392	9.580	1.500	8.368	7.534	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 16
Governing Disturbing Force : 13.014kN
Governing Restoring Force : 71.963kN
Minimum Sliding Ratio for the Critical Load Case : 5.529
Critical Load Case for Overturning about X-Direction : 17
Governing Overturning Moment : 19.181kNm
Governing Resisting Moment : 122.582kNm
Minimum Overturning Ratio for the Critical Load Case : 6.391

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 17

Governing Disturbing Force : 9.315kN

Governing Restoring Force : 70.237kN

Minimum Sliding Ratio for the Critical Load Case : 7.540

Critical Load Case for Overturning about Z-Direction : 16

Governing Overturning Moment : -27.548kNm

Governing Resisting Moment : 127.391kNm

Minimum Overturning Ratio for the Critical Load Case : 4.624

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 17

Governing Disturbing Force : 14.130kN

Governing Restoring Force : 99.331kN

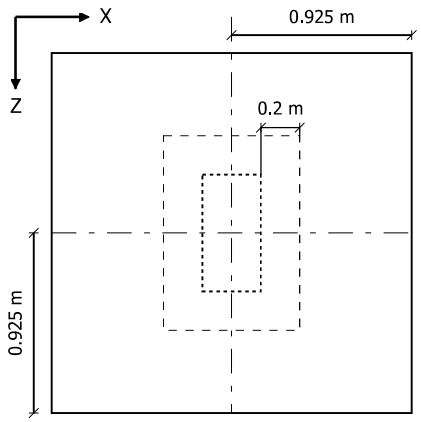
Minimum Sliding Ratio for the Critical Load Case : 7.030

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 176.515kN, Load Case # 22

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.176m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.805kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3398.657kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.805kN

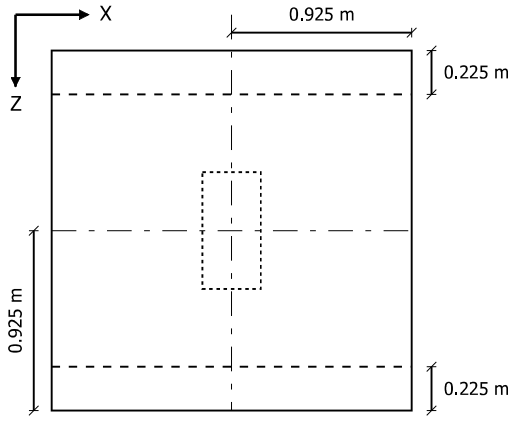
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1610.104kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 625.250kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.281m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

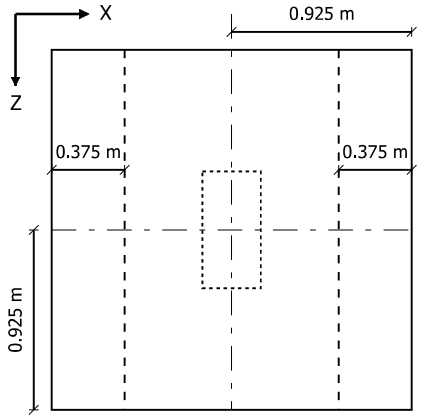
From above calculations, 0.75 X V_c = 468.938 kN

Critical load case for V_{ux} is # 26 $V_{ux} = V_{ux}|_{x=D_x}$ = 41.194 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 625.250 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.431 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

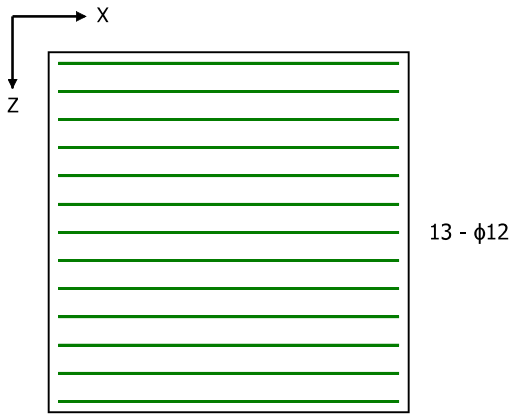
From above calculations, 0.75 X V_c = 468.938 kN

Critical load case for V_{uz} is # 27 $V_{uz} = V_{uz}|_{z=D_z}$ = 70.346 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 27

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl. 7.12.2, ρ_{min} =		0.00169	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.075	m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	47.958	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	53.286	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00058	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00050	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1251.620	mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 140.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover}$ = 0.725 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 13

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1470.255 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00231

From ACI Cl.7.6.1, minimum req'd clear distance between bars

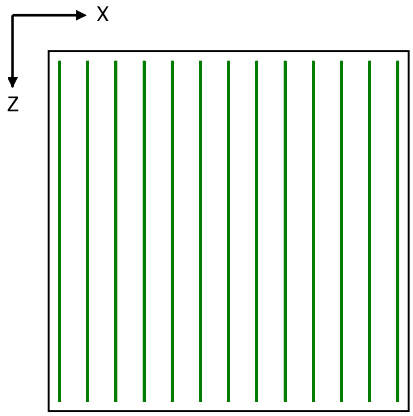
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 140.000mm

Provided Steel Area / Required Steel Area = 1.175

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



13 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 26

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00164

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.225 m

Ultimate moment, $M_u|_{x=D_x}$ = 27.954 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 31.060 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00036

(Based on gross depth) ρ x d_{eff} / Depth = 0.00030

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1216.698 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 140.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.575 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 13

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1470.255 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

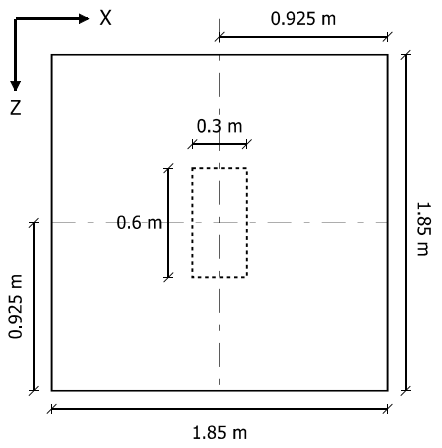
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00239

From ACI Cl.7.6.1, minimum req'd clear distance between bars

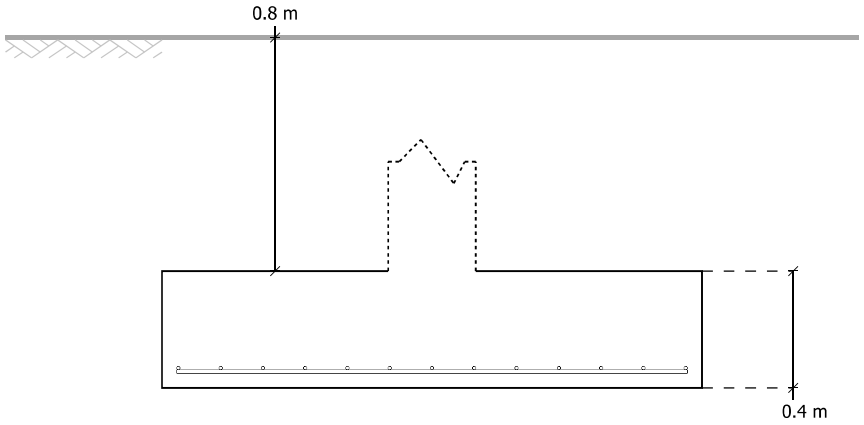
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 140.000mm

Provided Steel Area / Required Steel Area = 1.208

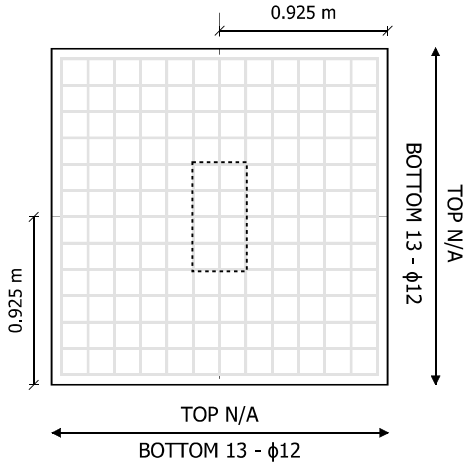
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

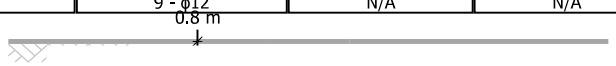
	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.85 m	1.85 m	0.4 m	N/A	N/A	13 - ϕ12	13 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

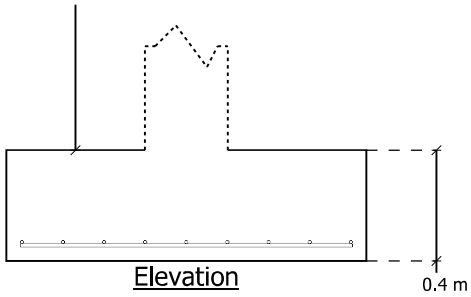
Isolated Footing Design(ACI 318-08)

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
78	1	1.300m	1.300m	0.400m

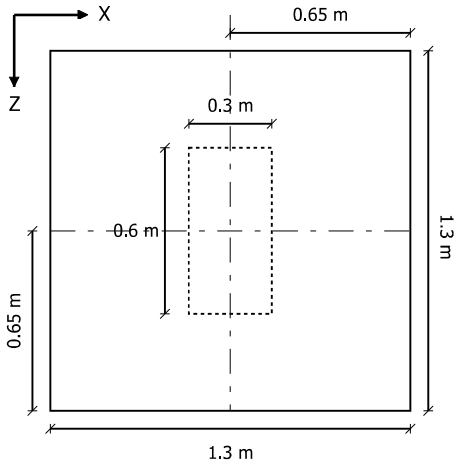
Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
78	9 - ϕ12	9 - ϕ12	N/A	N/A	N/A	N/A



FOUNDATION-F2 ALONG GRID 3 & G



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1300.000mm
Footing Width - Z (Fw) : 1300.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.300m

Column Width - Z (B_{col}) : 0.600m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 140.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.300m

Initial Width (W_o) = 1.300m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
8	SLS: (DL)	1.00	1.00	1.00
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
8	170.477	13.575	11.515	2.813	-5.918
9	181.325	14.421	5.665	2.535	-9.657
10	175.471	13.262	11.525	2.785	-5.843
11	174.223	13.340	11.522	2.792	-5.862
12	174.019	13.602	14.227	8.533	-5.956
13	174.235	17.909	11.511	2.811	-10.838
14	174.402	14.952	9.321	-0.423	-7.598
15	182.628	14.631	18.882	16.308	-7.512
16	102.525	10.294	3.974	-2.599	-5.866
17	113.494	9.866	16.722	19.709	-5.751
18	106.044	12.479	6.905	1.685	-8.470
19	105.828	8.172	9.621	7.408	-3.589
20	170.716	15.724	8.580	-1.474	-8.234
21	181.685	15.296	21.328	20.834	-8.119

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	238.668	19.005	16.121	3.938	-8.286
23	256.352	20.033	7.936	3.535	-13.483
24	256.352	20.033	7.936	3.535	-13.483

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
25	212.754	17.509	11.486	-0.098	-8.834
26	225.002	18.887	29.524	32.196	-10.585
27	207.452	19.572	9.127	-3.497	-10.769
28	225.002	18.887	29.524	32.196	-10.585
29	209.940	22.482	13.812	3.372	-14.130
30	209.632	16.328	17.693	11.547	-7.156
31	153.812	15.656	5.667	-4.327	-9.031
32	171.362	14.971	26.064	31.366	-8.847
33	158.489	12.256	14.238	10.703	-5.380
34	158.797	18.409	10.358	2.528	-12.354

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 1.690 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.133 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.300 m Governing Load Case : # 8

Width (W_2) = 1.300 m Governing Load Case : # 8

Depth (D_2) = 0.400 m Governing Load Case : # 26

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 1.690 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 16.223 kN

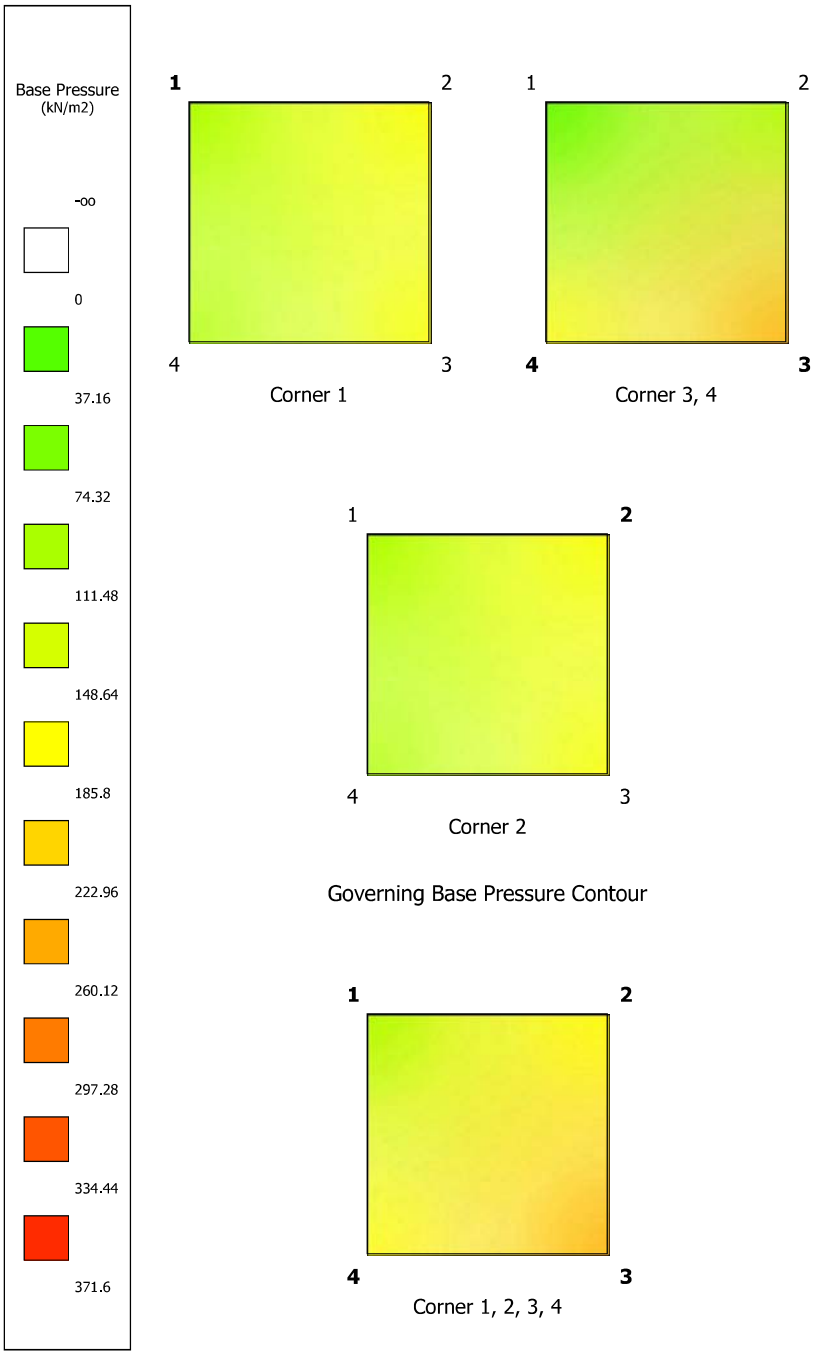
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 21.743 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

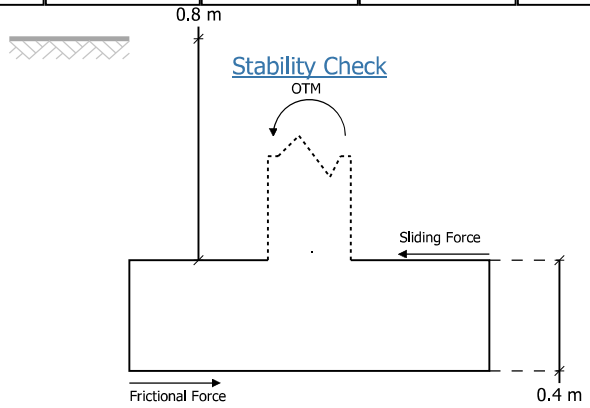


Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
14	**79.5495**	153.7193	171.7740	97.6042	0.000
9	74.5187	**158.7753**	184.9979	100.7413	0.000
21	10.8919	88.6551	**249.0508**	171.2876	0.000
21	10.8919	88.6551	249.0508	**171.2876**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
14	**79.5495**	153.7193	171.7740	97.6042
9	74.5187	**158.7753**	184.9979	100.7413
21	10.8919	88.6551	**249.0508**	171.2876
21	10.8919	88.6551	249.0508	**171.2876**



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
8	8.138	9.594	8.777	1.500	18.262	12.050	1.500
9	8.059	20.516	10.609	1.500	29.690	9.322	1.500
10	8.530	9.816	9.105	1.500	18.760	12.558	1.500
11	8.430	9.760	9.022	1.500	18.635	12.430	1.500
12	8.260	7.897	8.072	1.500	9.687	12.201	1.500
13	6.280	9.770	7.471	1.500	18.602	7.732	1.500
14	7.528	12.075	9.034	1.500	41.760	10.259	1.500
15	7.991	6.192	6.922	1.500	6.009	10.823	1.500
16	7.233	18.737	9.543	1.500	90.463	9.273	1.500
17	8.137	4.801	5.847	1.500	3.729	10.282	1.500
18	6.116	11.054	7.568	1.500	21.048	7.047	1.500
19	9.326	7.921	8.538	1.500	8.303	13.814	1.500
20	7.034	12.891	8.732	1.500	69.272	9.427	1.500
21	7.611	5.458	6.273	1.500	4.862	10.117	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 18
Governing Disturbing Force : 12.479kN
Governing Restoring Force : 76.326kN
Minimum Sliding Ratio for the Critical Load Case : 6.116
Critical Load Case for Overturning about X-Direction : 17
Governing Overturning Moment : 26.398kNm
Governing Resisting Moment : 98.448kNm
Minimum Overturning Ratio for the Critical Load Case : 3.729

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 17

Governing Disturbing Force : 16.722kN

Governing Restoring Force : 80.274kN

Minimum Sliding Ratio for the Critical Load Case : 4.801

Critical Load Case for Overturning about Z-Direction : 18

Governing Overturning Moment : -13.462kNm

Governing Resisting Moment : 94.869kNm

Minimum Overturning Ratio for the Critical Load Case : 7.047

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 17

Governing Disturbing Force : 19.415kN

Governing Restoring Force : 113.525kN

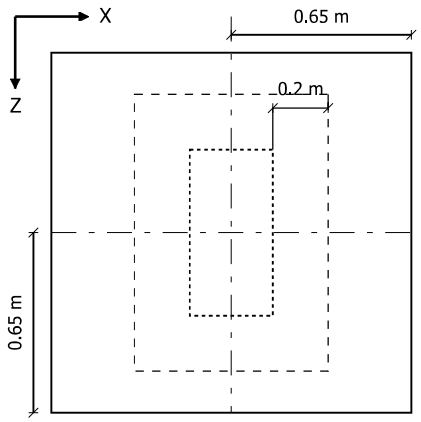
Minimum Sliding Ratio for the Critical Load Case : 5.847

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 188.445kN, Load Case # 23

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.176m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.805kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3398.657kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.805kN

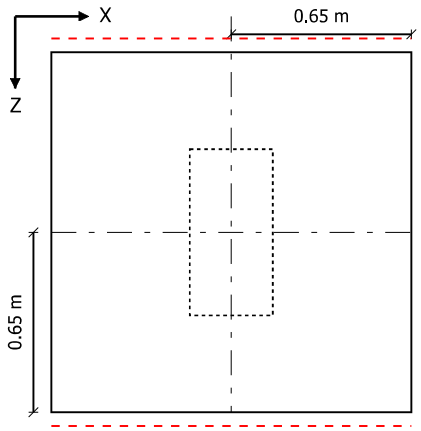
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1610.104kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 439.365kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.006m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

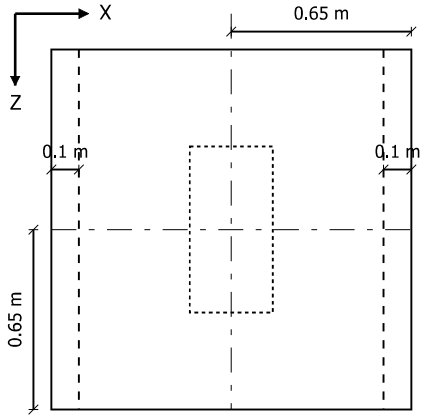
From above calculations, 0.75 X V_c = 329.524 kN

Critical load case for V_{ux} is # 26 $V_{ux} = V_{ux}|_{x=D_x}$ = 2.118 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 439.365 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.156 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

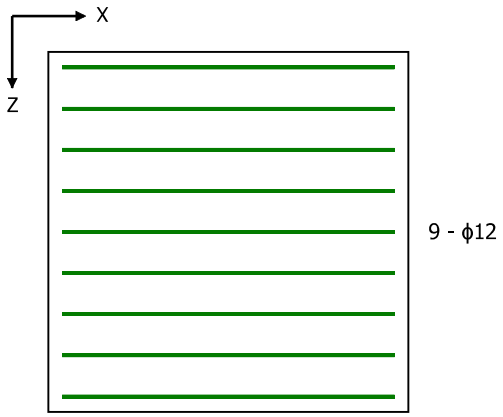
From above calculations, 0.75 X V_c = 329.524 kN

Critical load case for V_{uz} is # 23 $V_{uz} = V_{uz}|_{z=D_z}$ = 45.795 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 23

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl. 7.12.2, ρ_{min} =		0.00177	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{(0.85 \times F_c')}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.800	m
Ultimate moment,	$M_u\big\vert_{z=D_x}$ =	35.401	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	39.334	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{(F_y \times W \times d_{eff}^2)}}\right]$ =	0.00061	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00053	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	918.298	mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 140.000mm

Selected spacing (S) = 148.500mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 145mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover}$ = 0.450 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00228

From ACI Cl.7.6.1, minimum req'd clear distance between bars

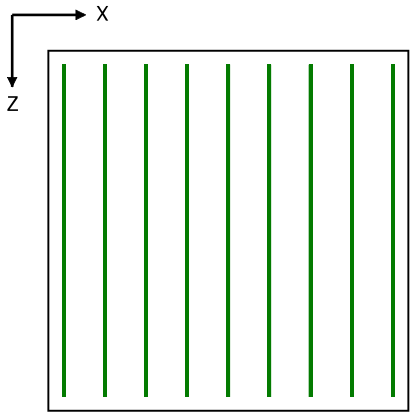
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 140.000mm

Provided Steel Area / Required Steel Area = 1.108

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



9 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 26

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00174

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.950 m

Ultimate moment, $M_u|_{x=D_x}$ = 20.297 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 22.552 kNm

(Based on effective depth) Required $\rho = \frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00038

(Based on gross depth) ρ x d_{eff} / Depth = 0.00031

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 902.299 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 140.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.300 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

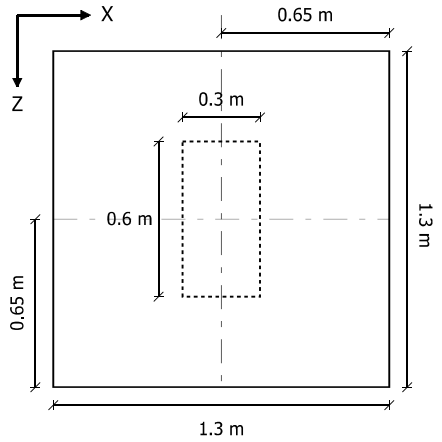
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00236

From ACI Cl.7.6.1, minimum req'd clear distance between bars

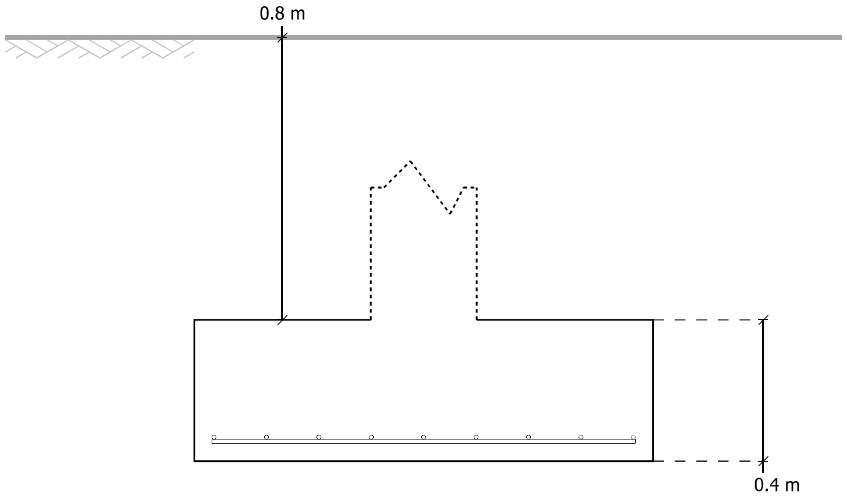
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 140.000mm

Provided Steel Area / Required Steel Area = 1.128

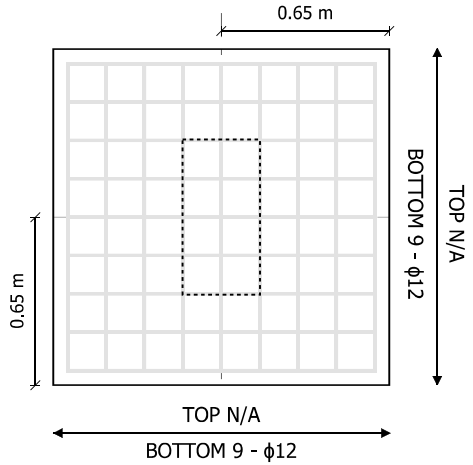
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.3 m	1.3 m	0.4 m	N/A	N/A	9 - ϕ12	9 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

ATTACHMENT 3 COMMENT RESOLUTION LOG SHEET

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Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT:	**MANSOURAH MASSARAH GOLD PROJECT**	**PROJECT NO.: 216**
STAGE:	**4 - PROJECT EXECUTION.**	**REVIEW LOG DOC. NO. & REV.:**
ACONEX REF. NO. & DATE:	BECHTEL-WTRAN-001510, 02 March 2020	

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 02 March 2020	Comment by	EPC Response DATE: 11 March,2020	Status	Remarks
			General Comments (cc)				
MD-216-6000-EG-CV-CAL-1480-A01	6000	1	Please review if you require these beam for improving lateral stiffness of the building .	Mr.Klaus Grof (BECHTEL)	As per Table-10.12, SBC-301-2007, the Allowable Drift is H/100 i.,e 33mm where as attained values as per Analysis is 25mm (Refer Page-17 in MD-216-6000-EG-CV-CAL-1480-C01 for details) . Additionally, the HCS Slab diaphragm action with Concrete screed & extra joint reinforcement provided will further restrain the Building laterally. So the Secondary Beams can be neglected & will not be required.	CLOSED	